

pearl_kuuridong_SR2

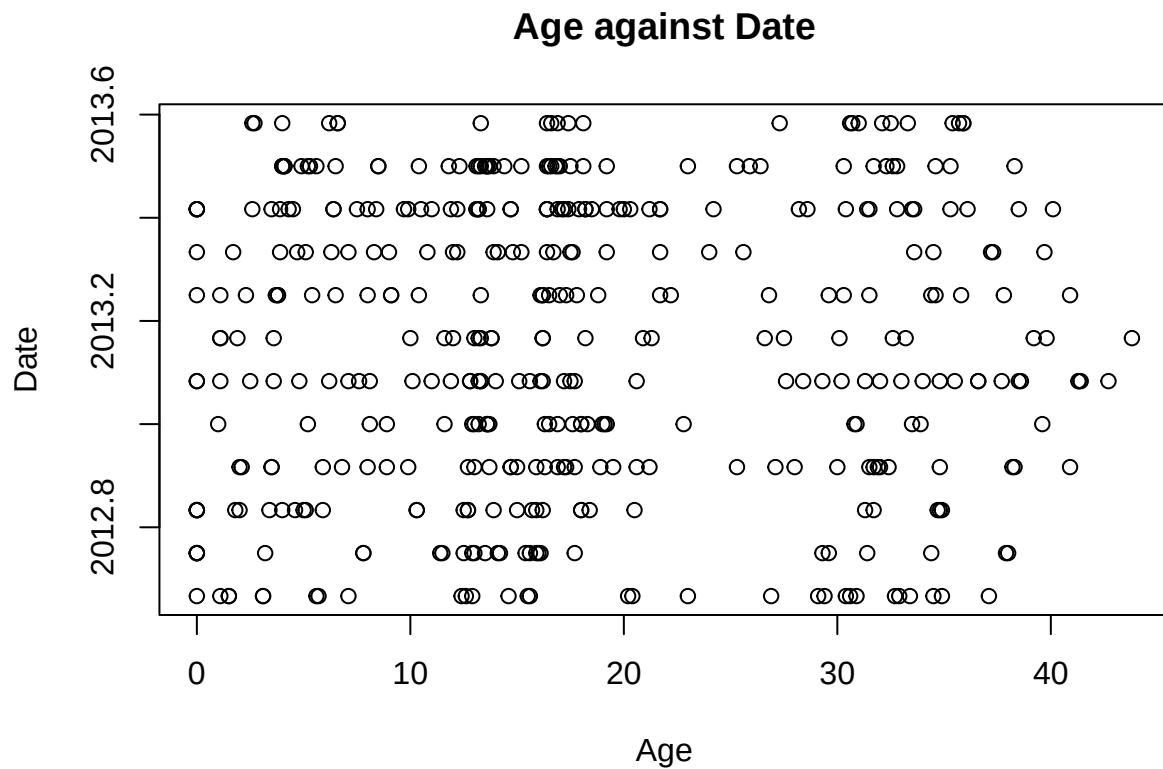
pearl_kuuridong_SR2

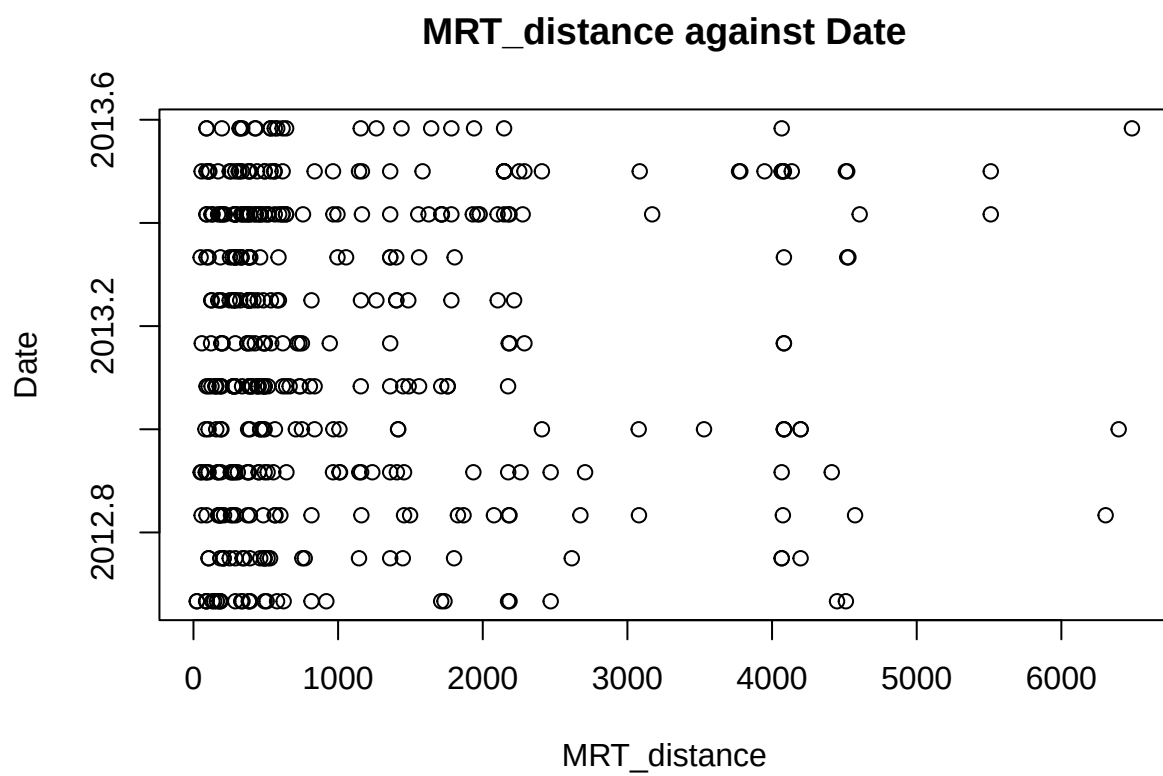
2025-11-20

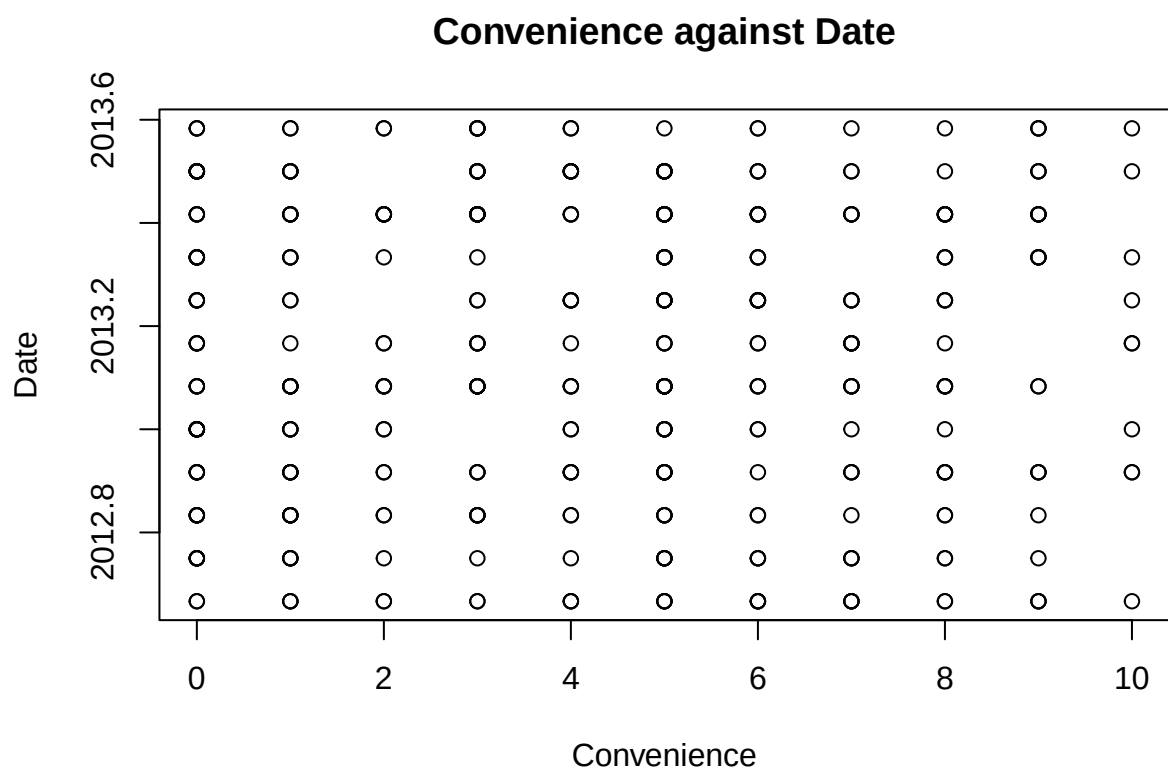
```
## # A tibble: 6 x 7
##   Date   Age MRT_distance Convenience Latitude Longitude Price
##   <dbl> <dbl>         <dbl>         <dbl>    <dbl>    <dbl> <dbl>
## 1 2013.   32          84.9           10     25.0     122.  37.9
## 2 2013.   19.5        307.           9      25.0     122.  42.2
## 3 2014.   13.3        562.           5      25.0     122.  47.3
## 4 2014.   13.3        562.           5      25.0     122.  54.8
## 5 2013.    5         391.           5      25.0     122.  43.1
## 6 2013.    7.1       2175.           3      25.0     122.  32.1
```

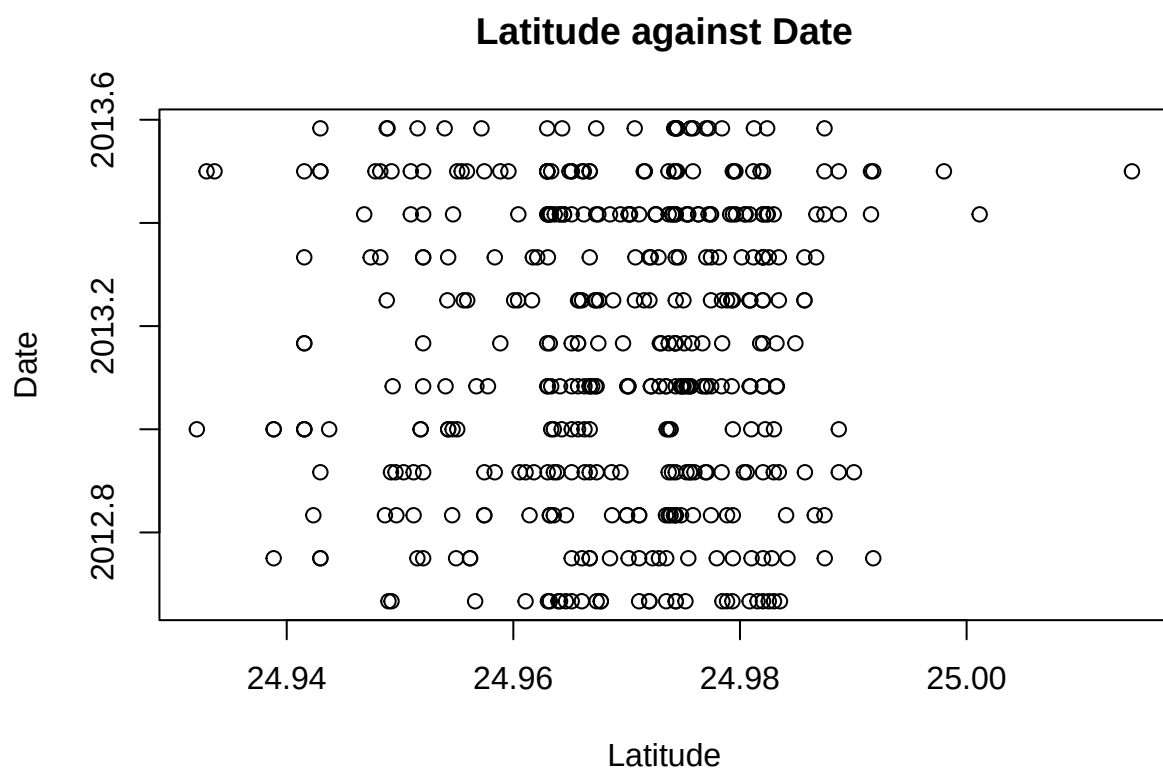
Question One

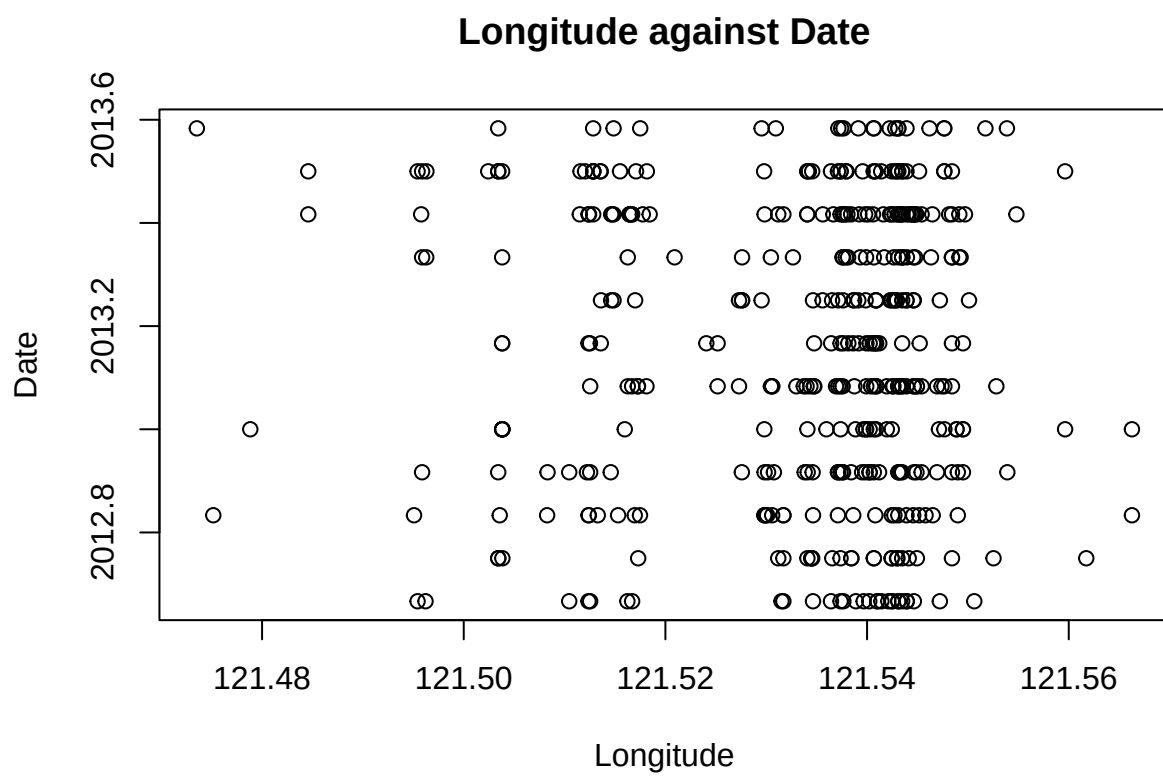
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## corrplot 0.95 loaded
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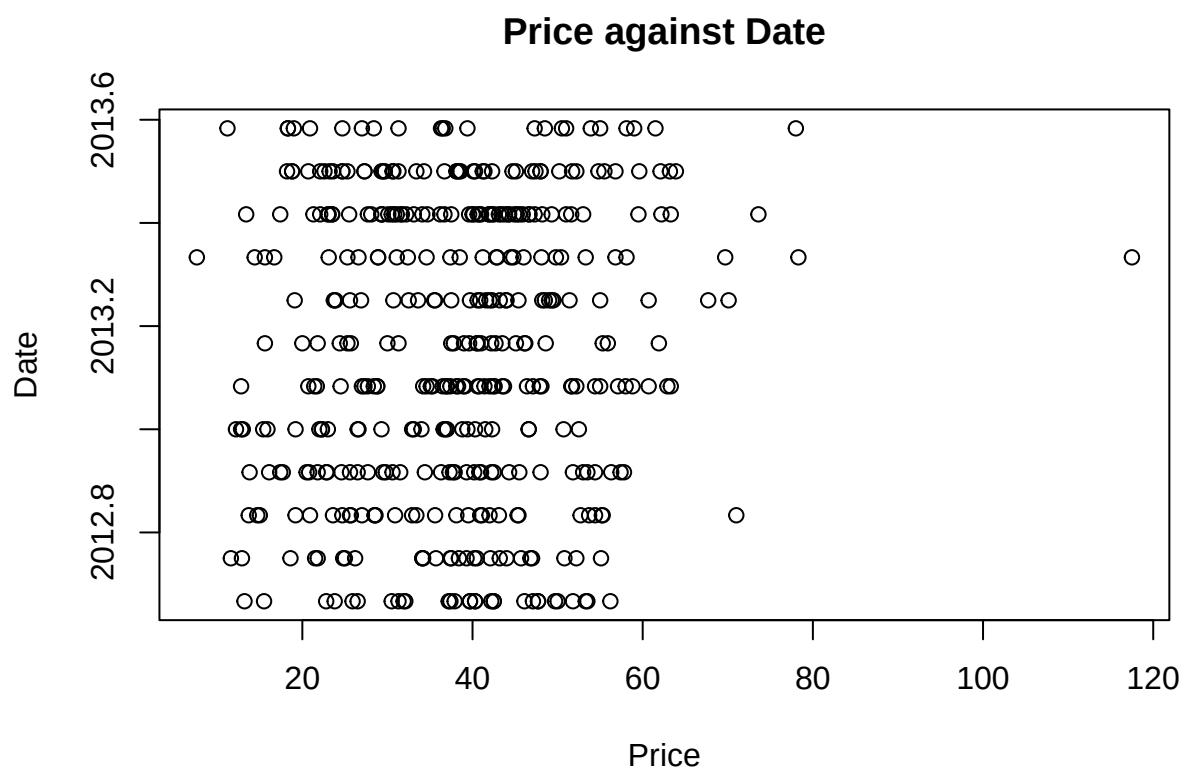


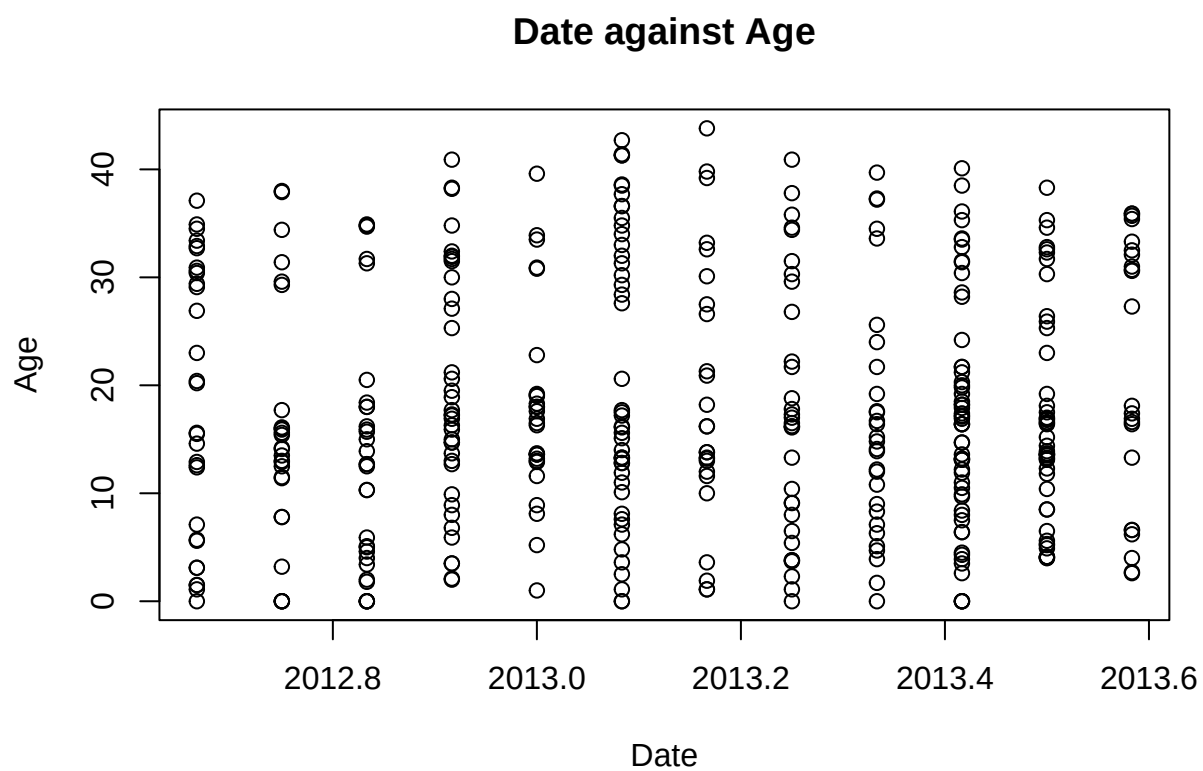




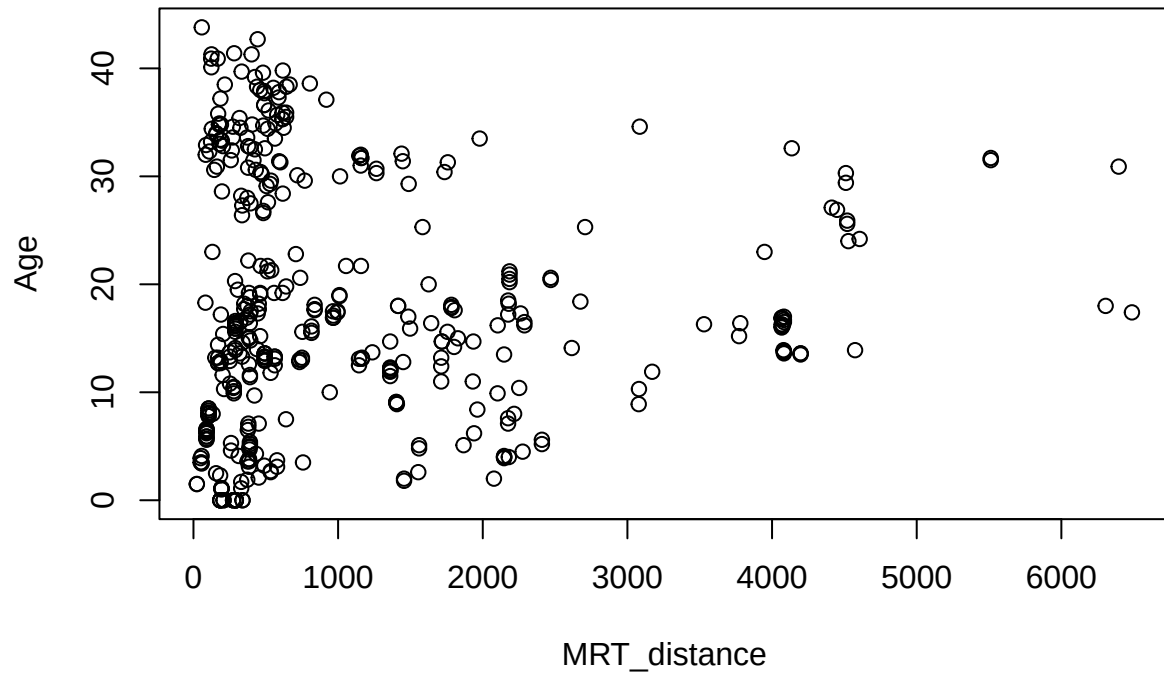




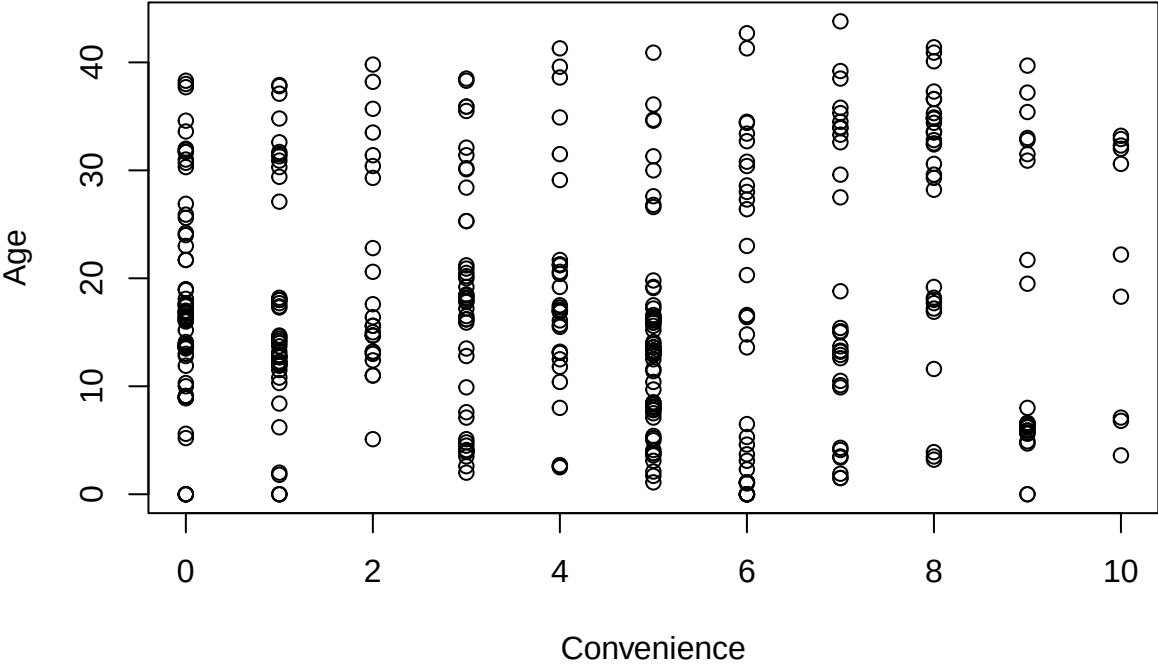




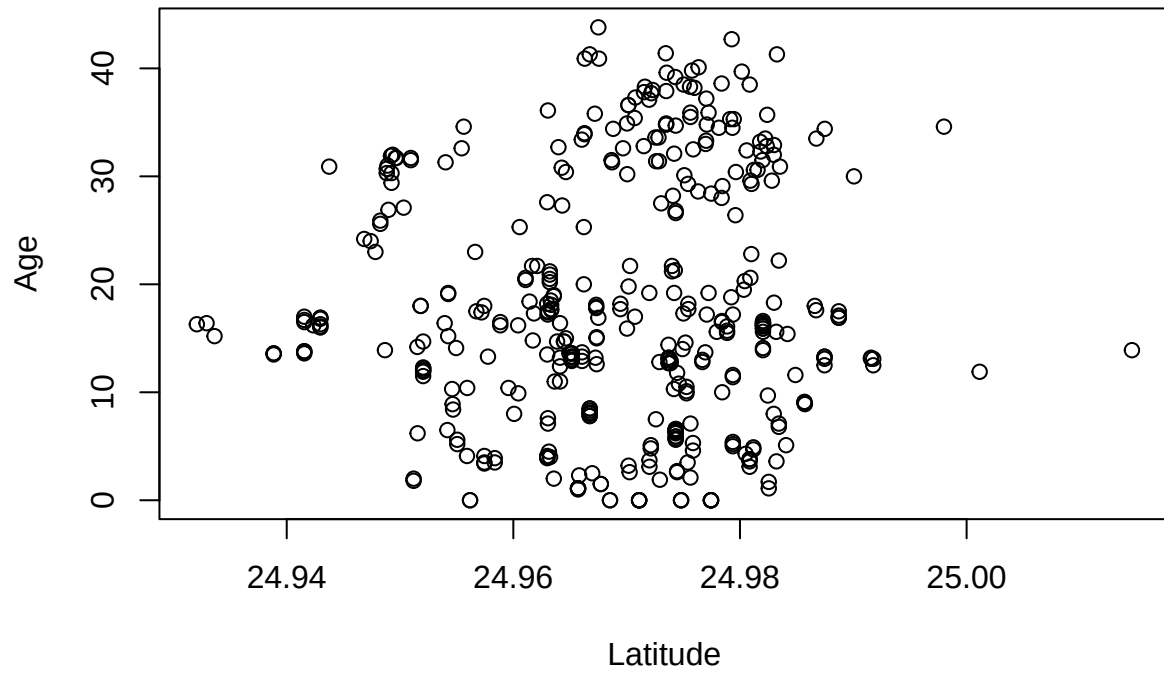
MRT_distance against Age



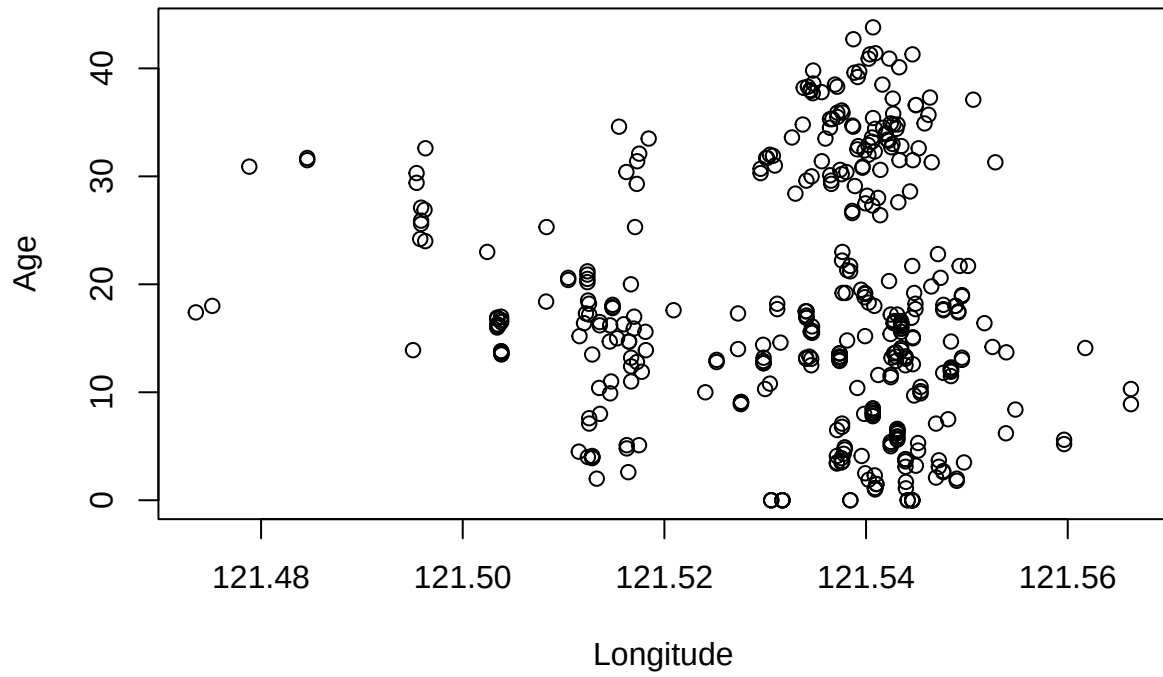
Convenience against Age



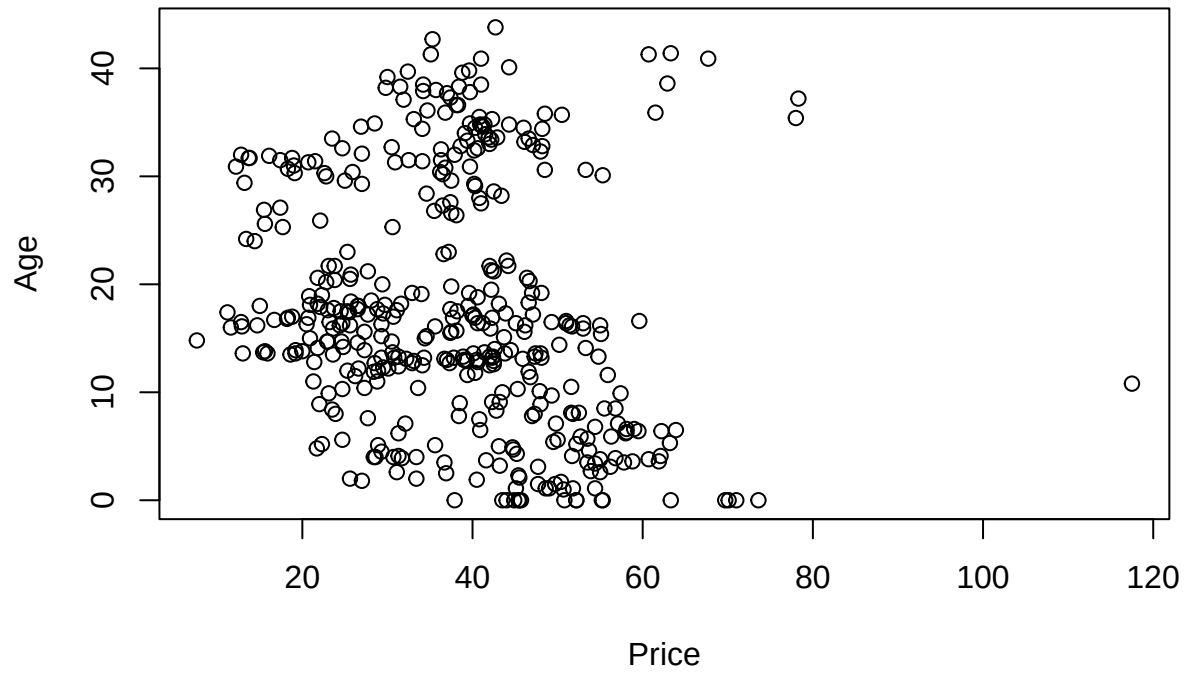
Latitude against Age



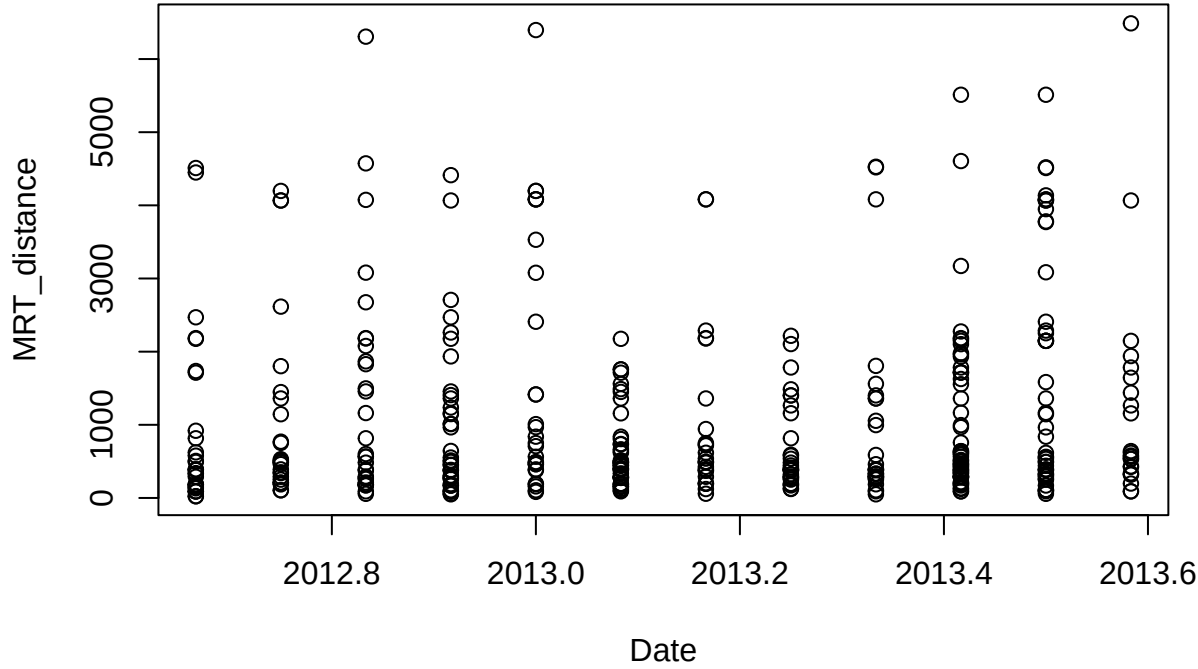
Longitude against Age



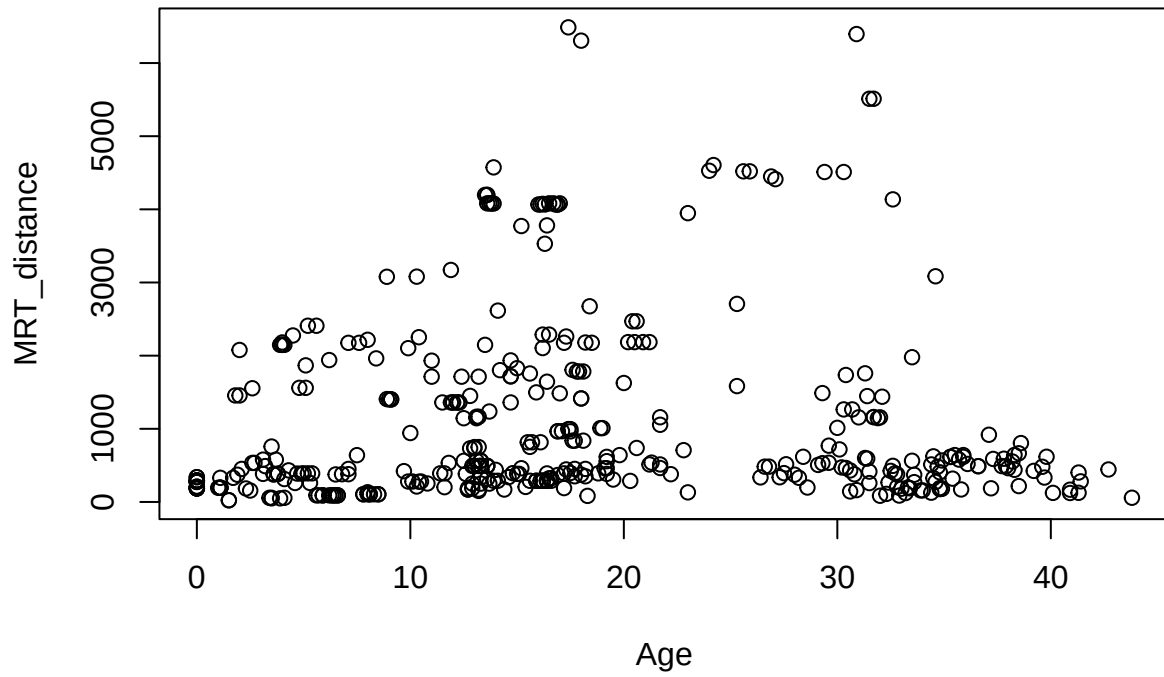
Price against Age



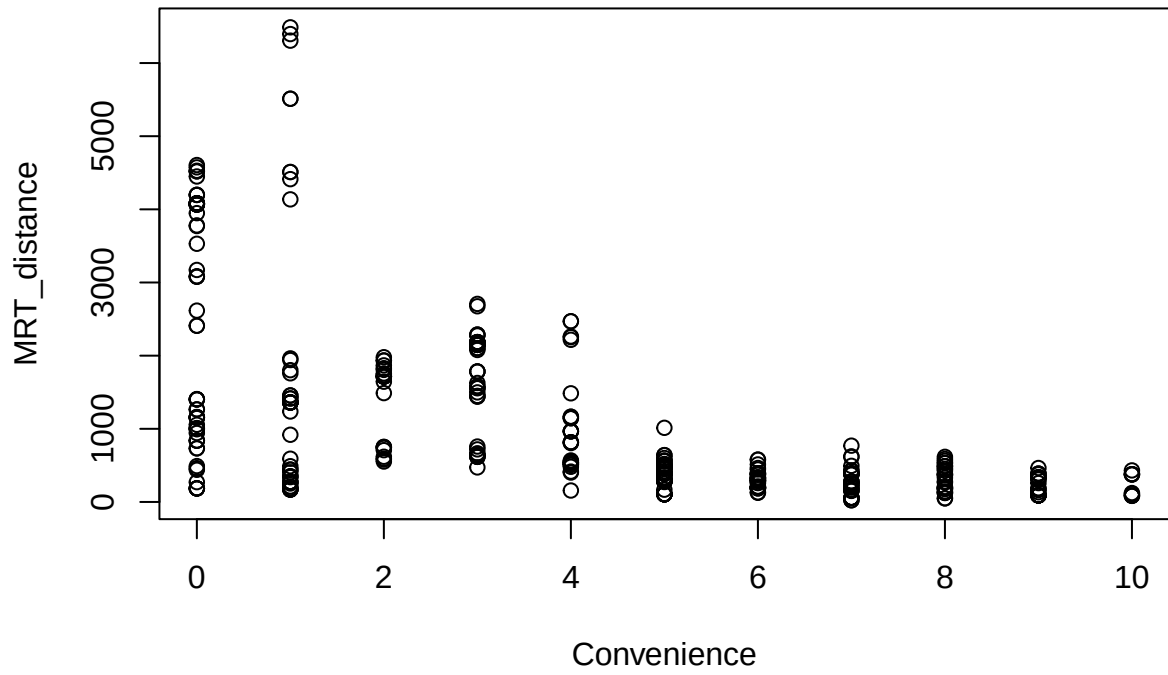
Date against MRT_distance



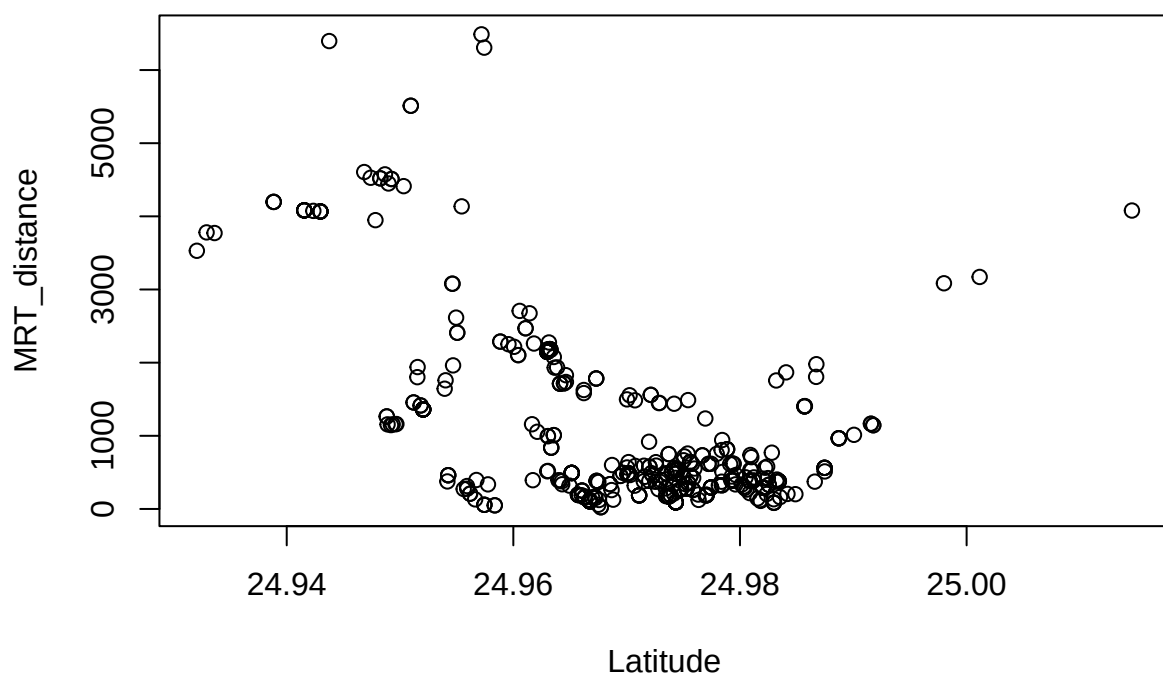
Age against MRT_distance



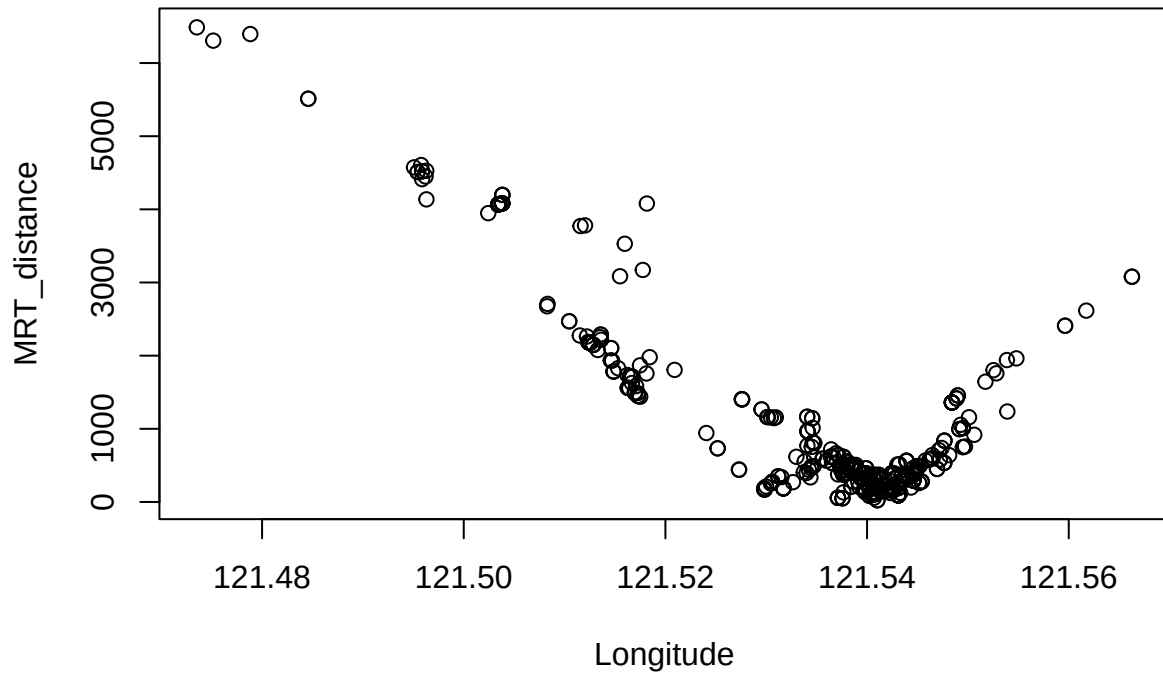
Convenience against MRT_distance



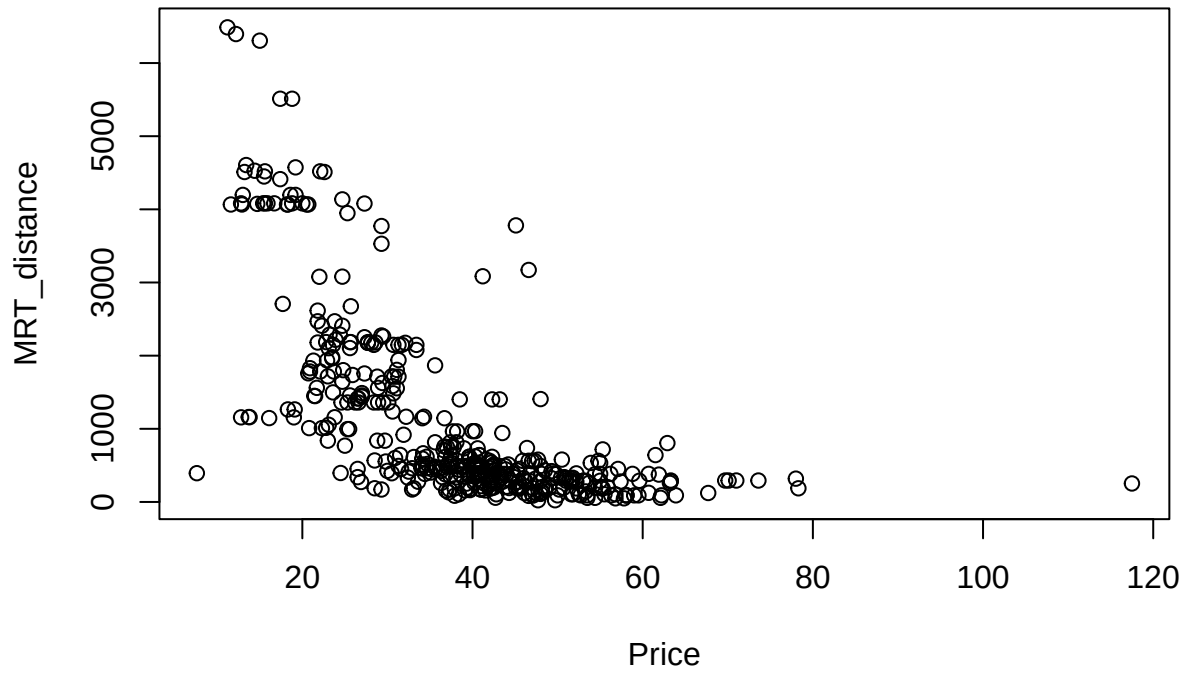
Latitude against MRT_distance



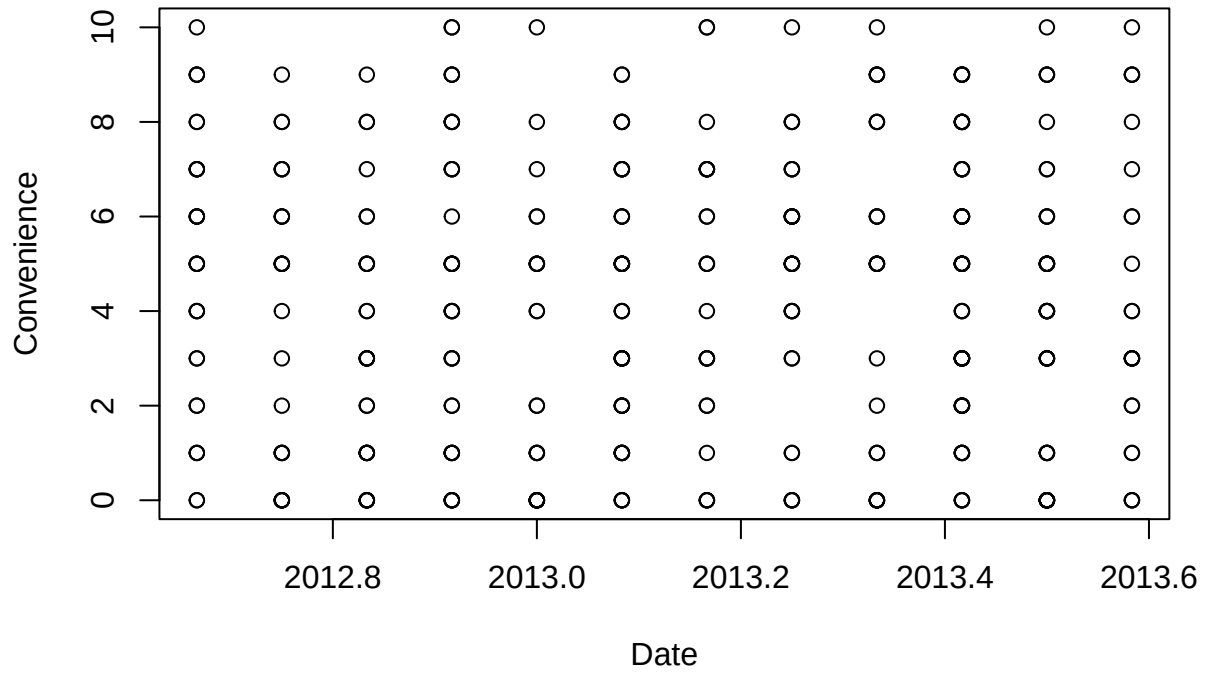
Longitude against MRT_distance



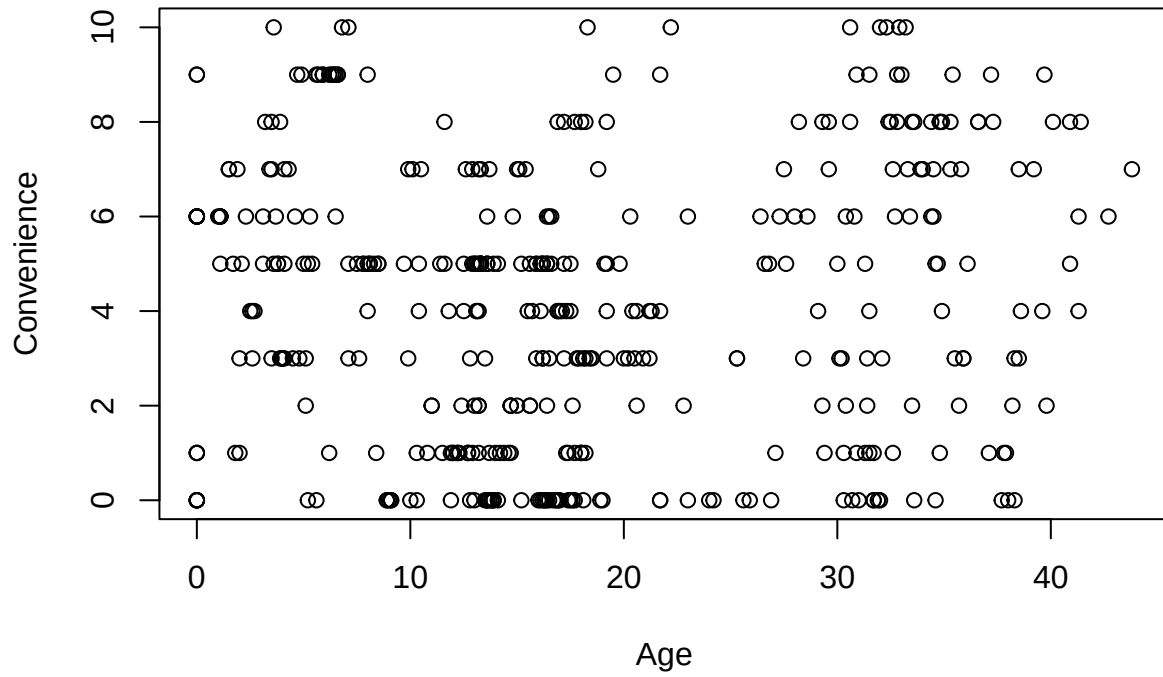
Price against MRT_distance



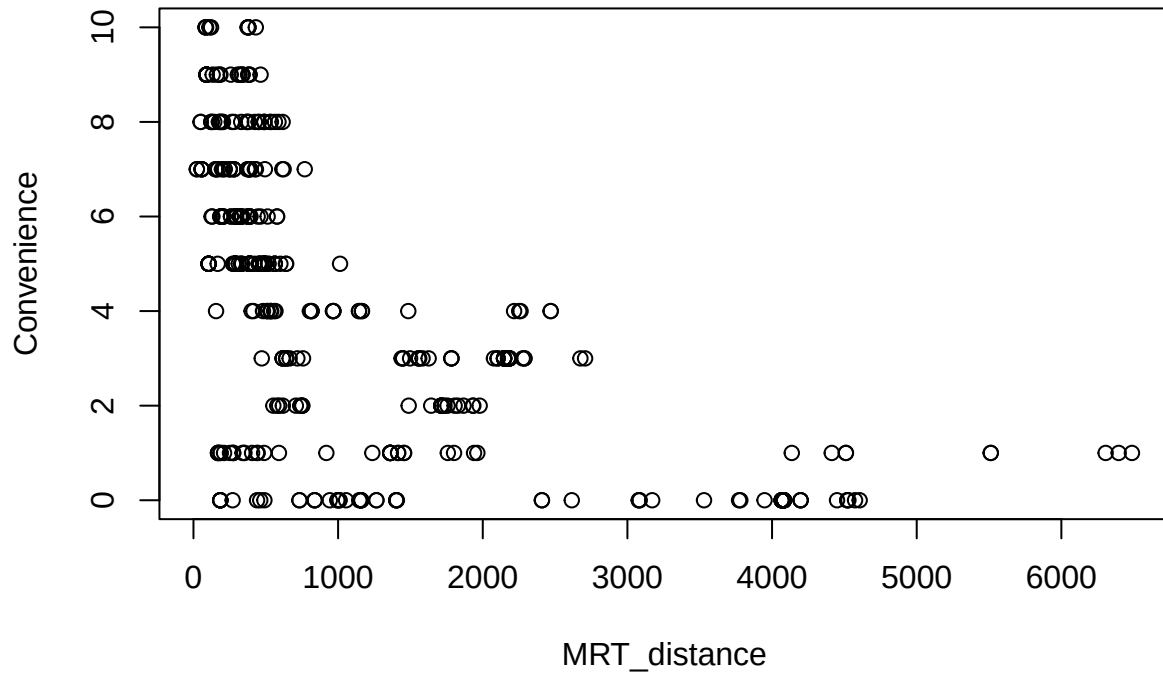
Date against Convenience



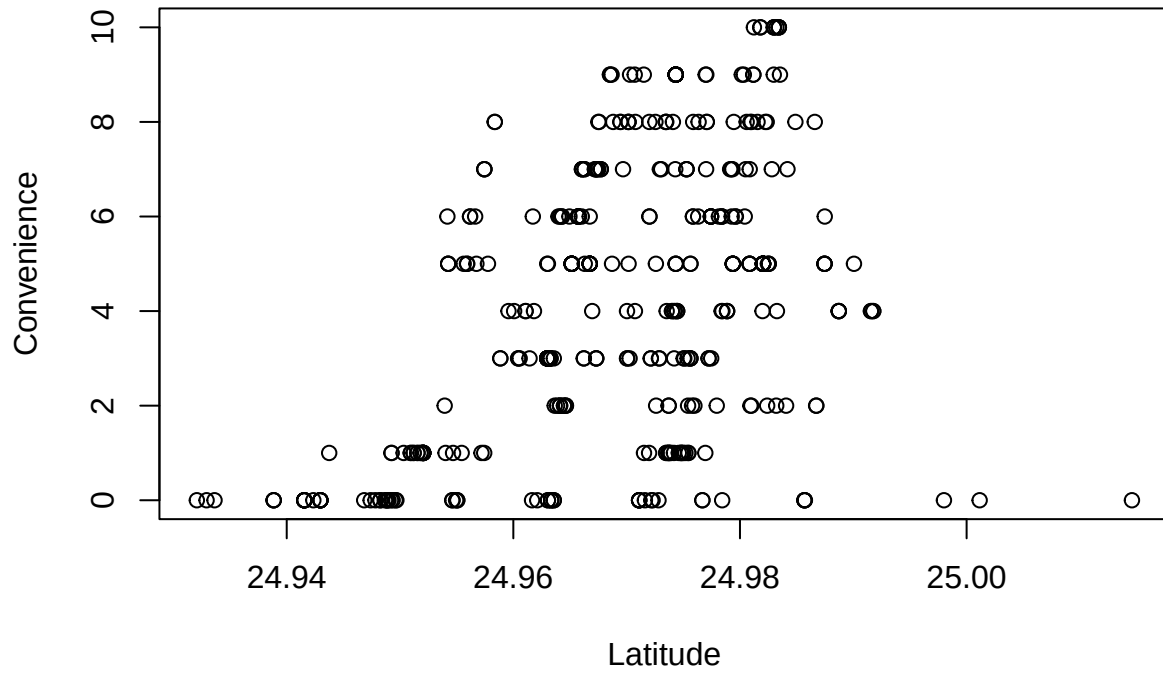
Age against Convenience



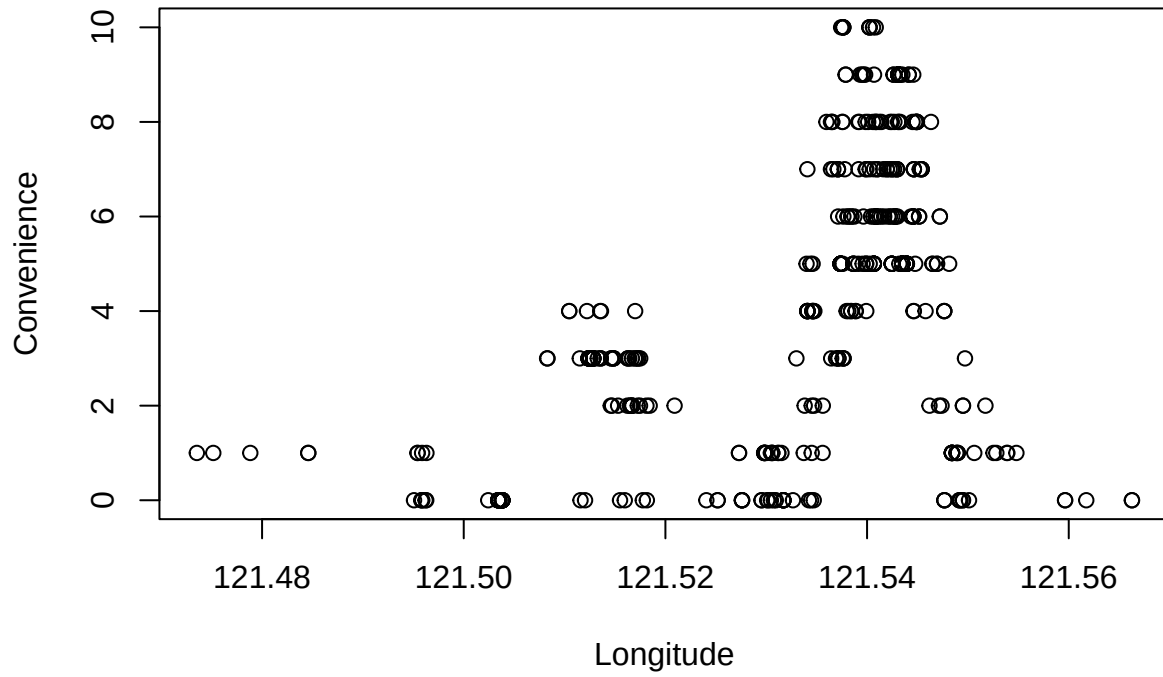
MRT_distance against Convenience



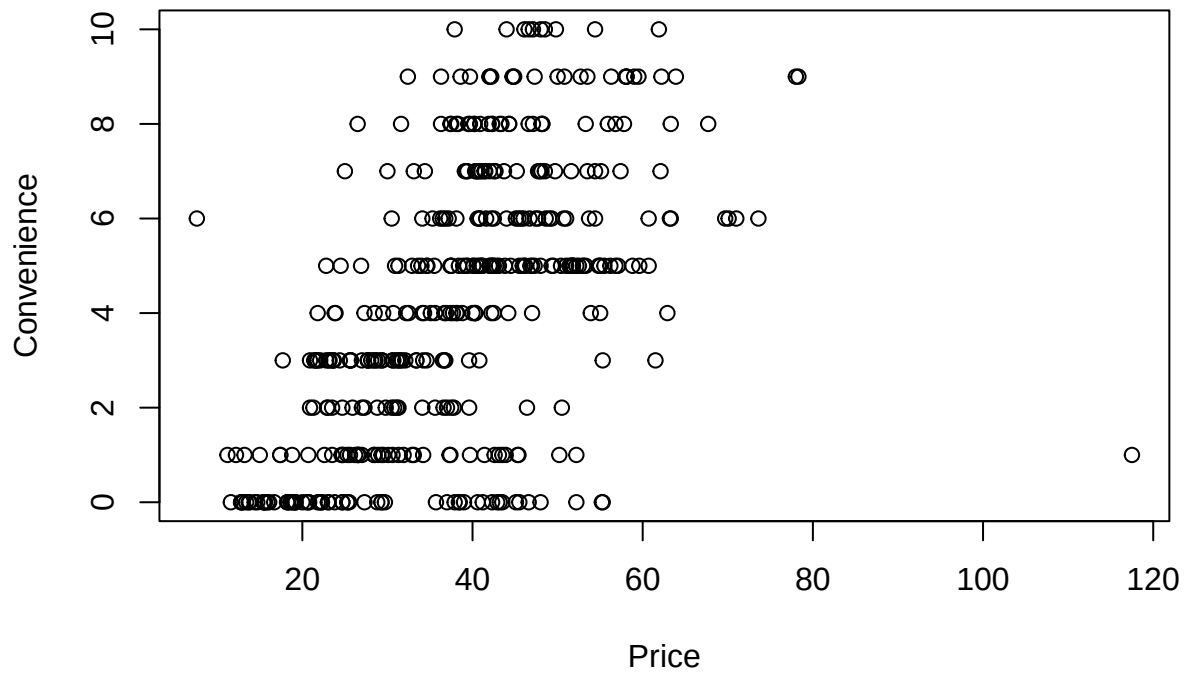
Latitude against Convenience

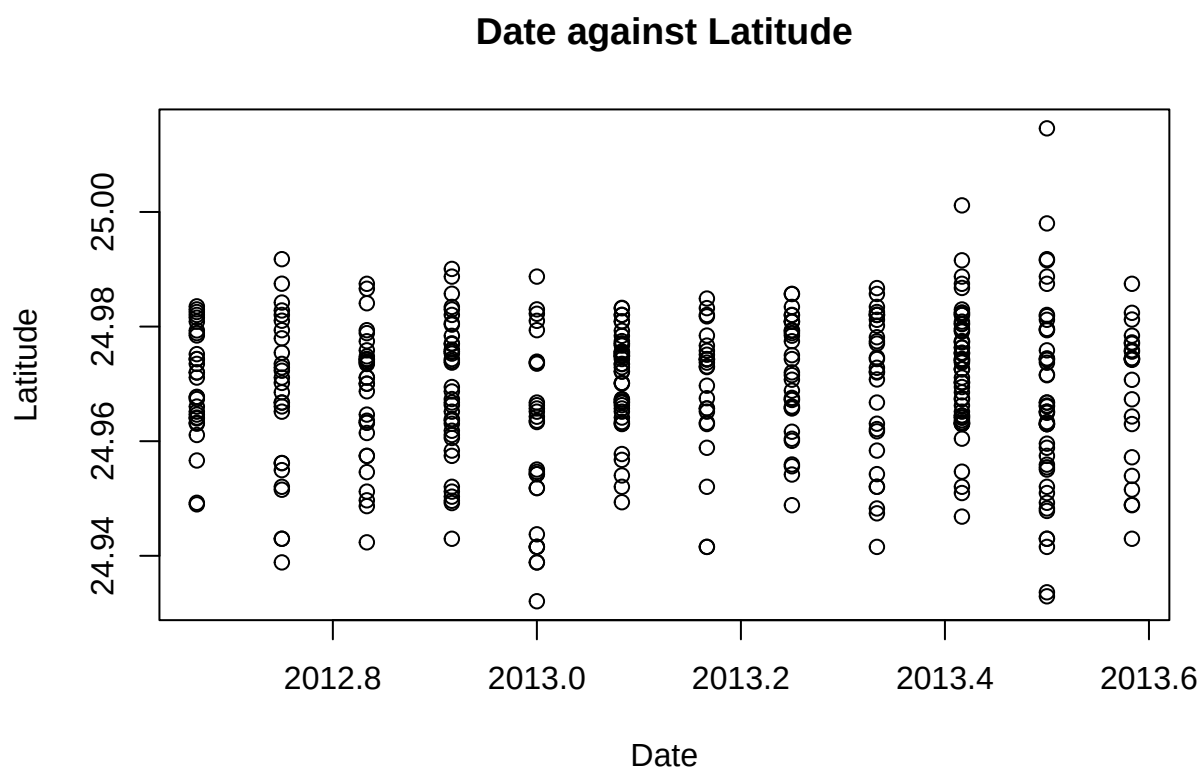


Longitude against Convenience

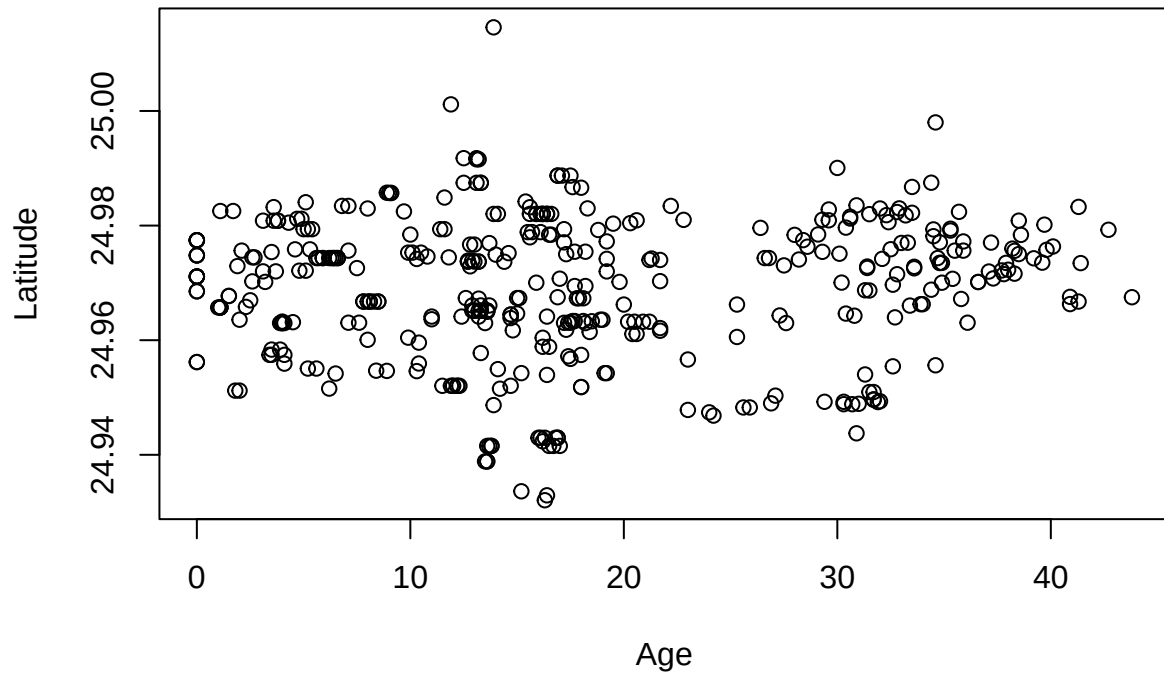


Price against Convenience

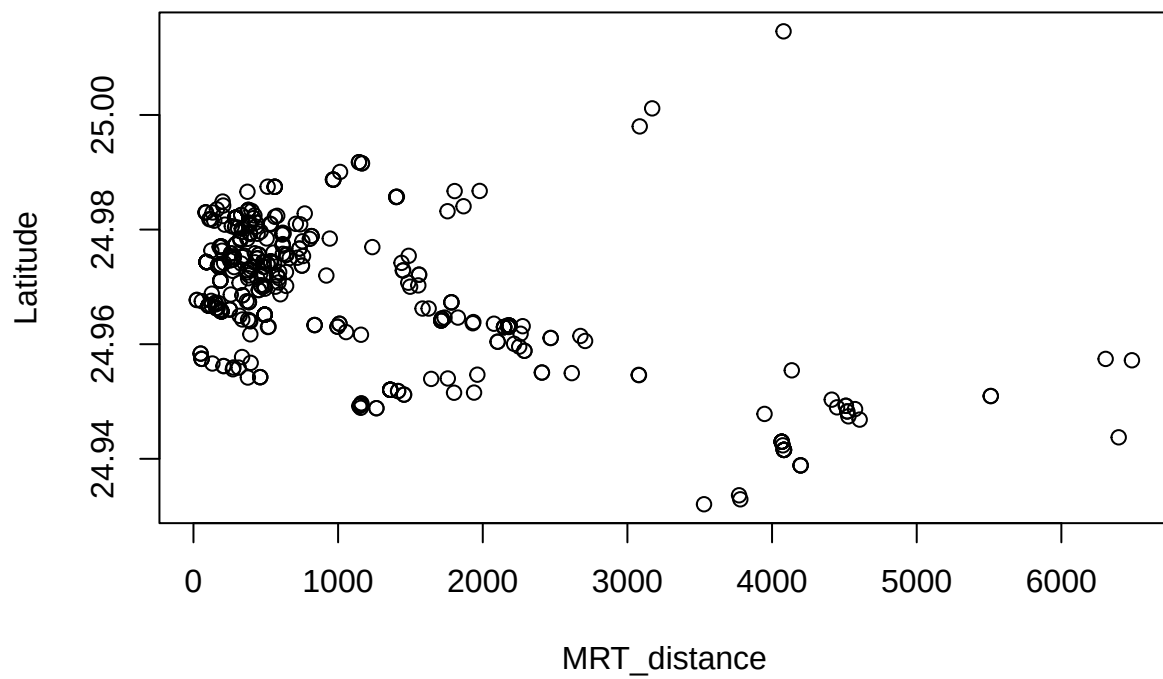




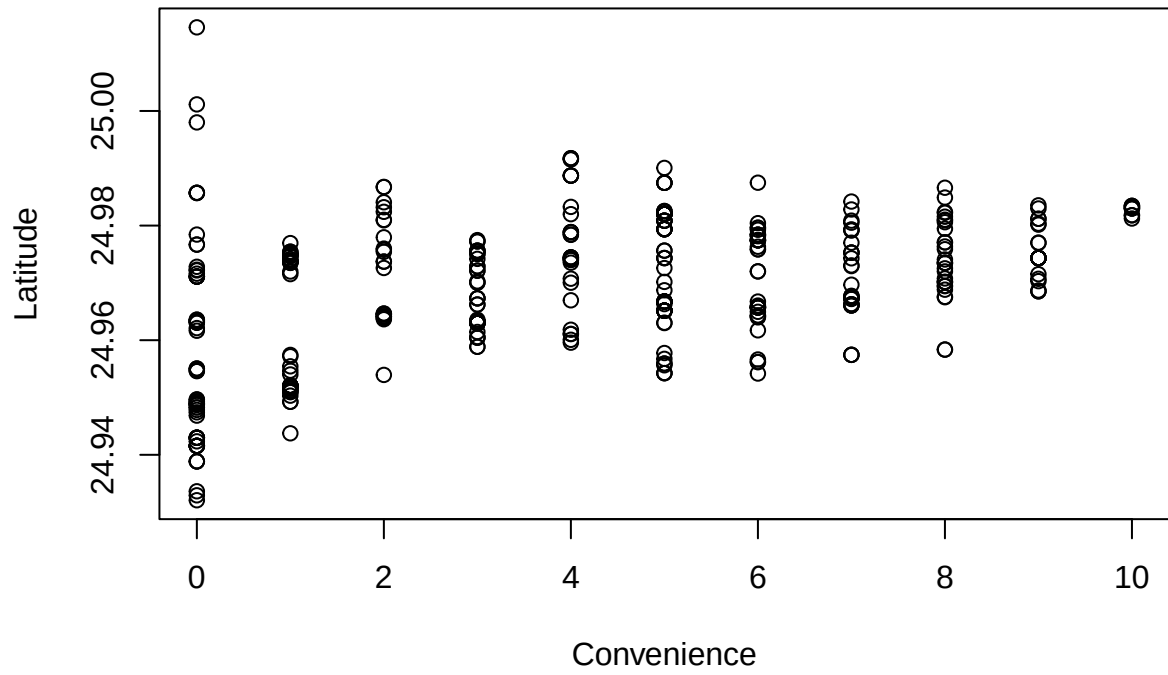
Age against Latitude



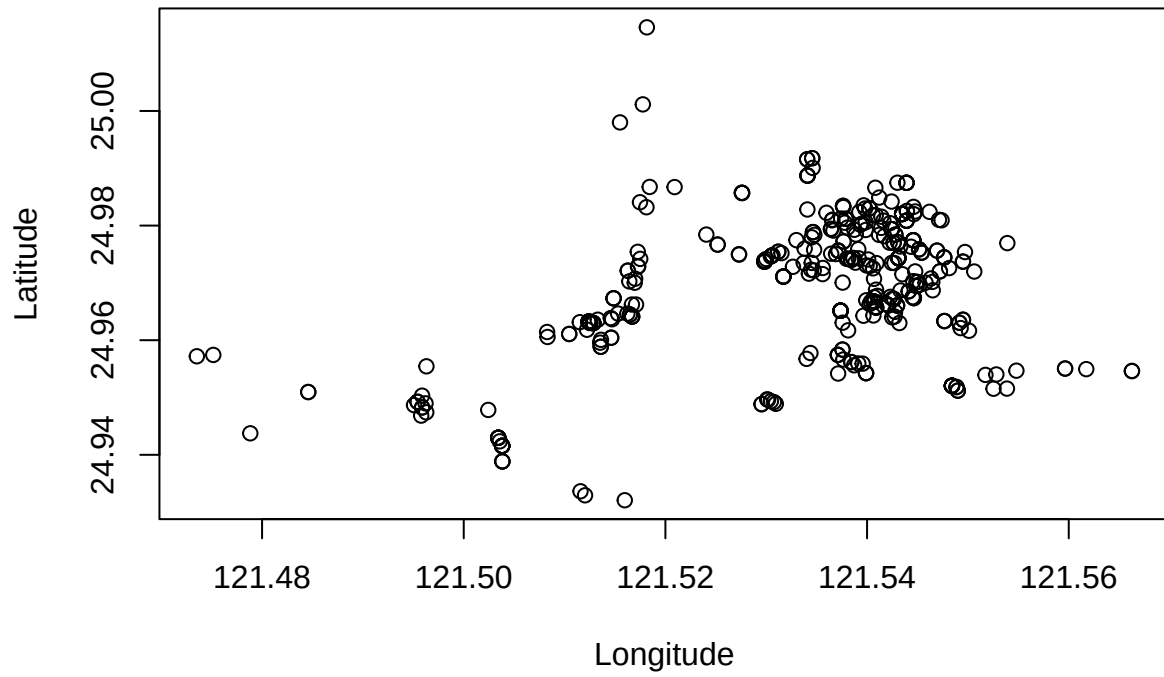
MRT_distance against Latitude

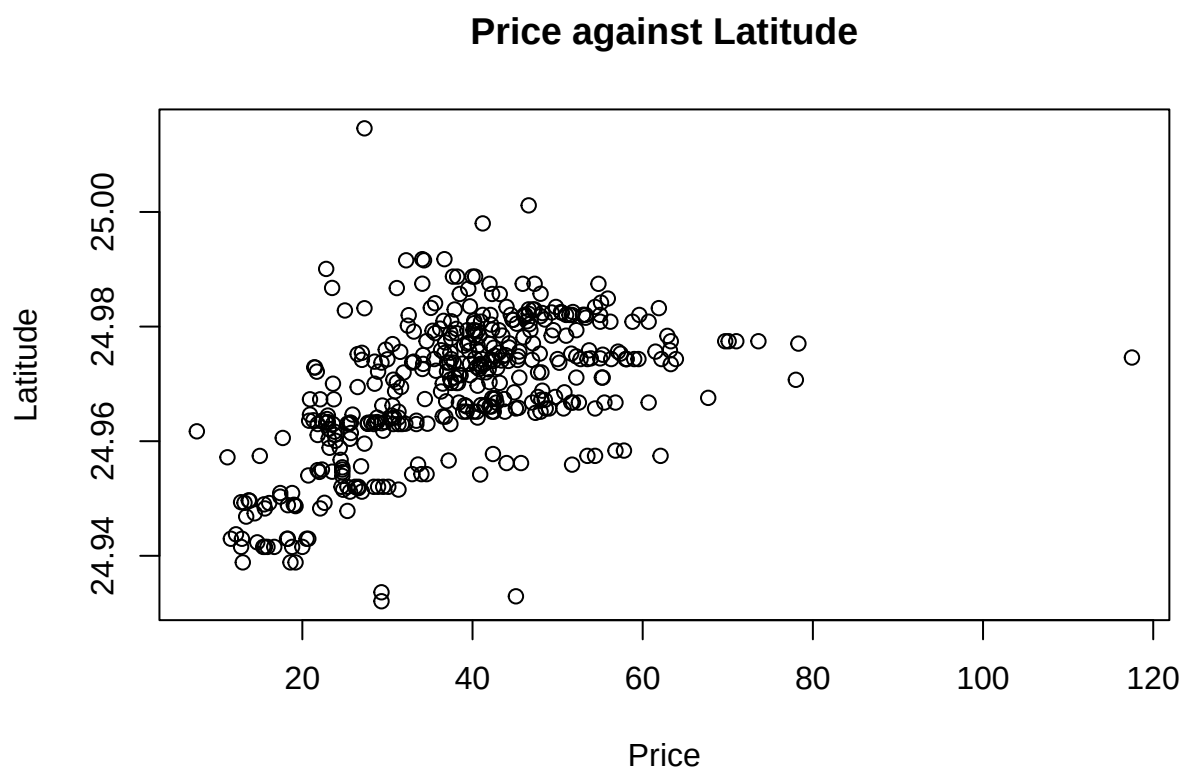


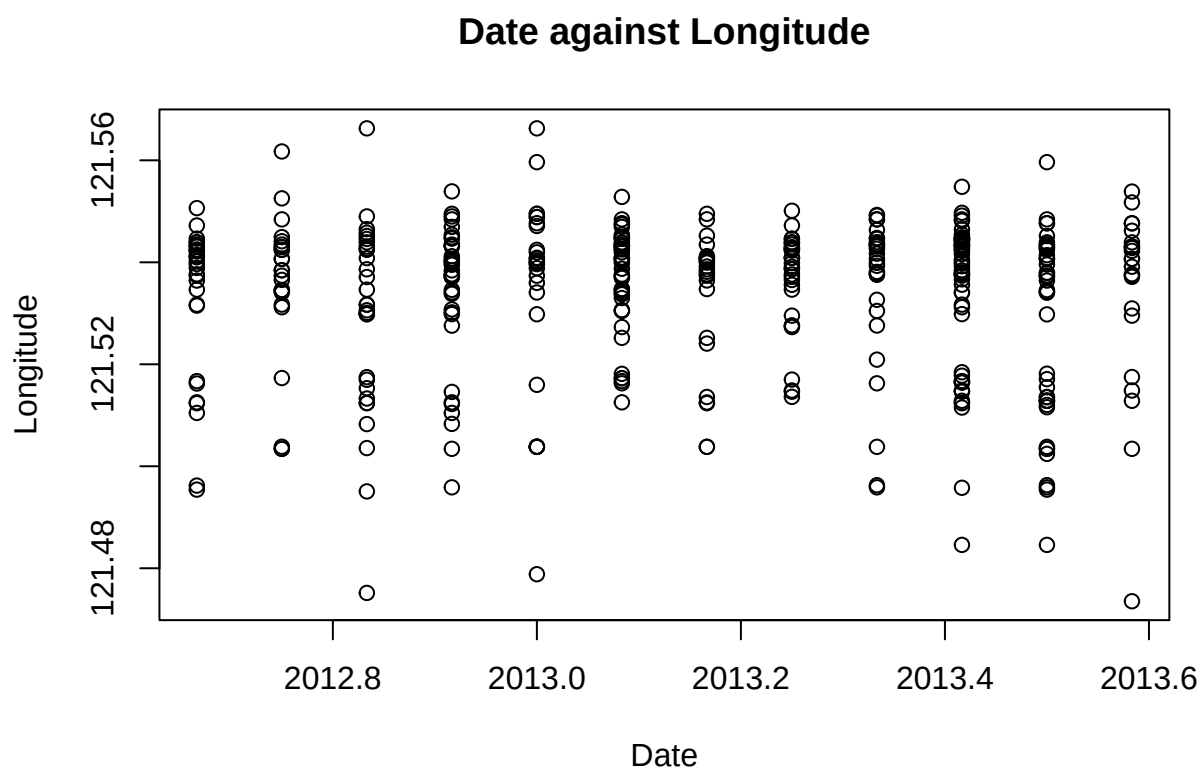
Convenience against Latitude



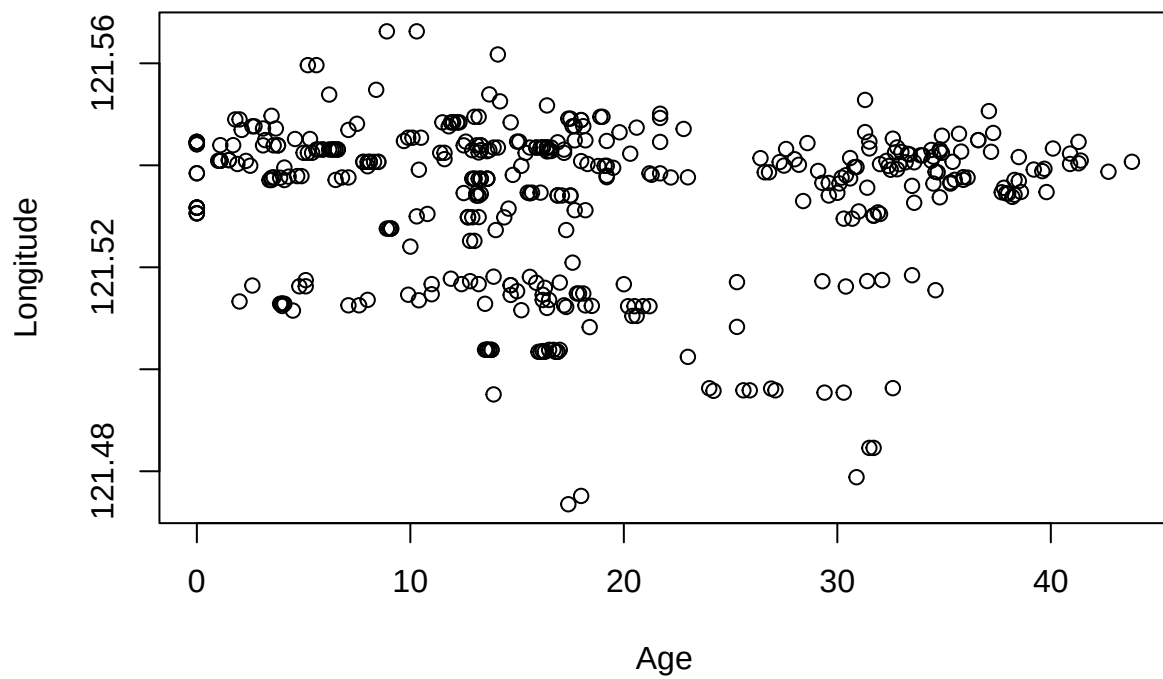
Longitude against Latitude



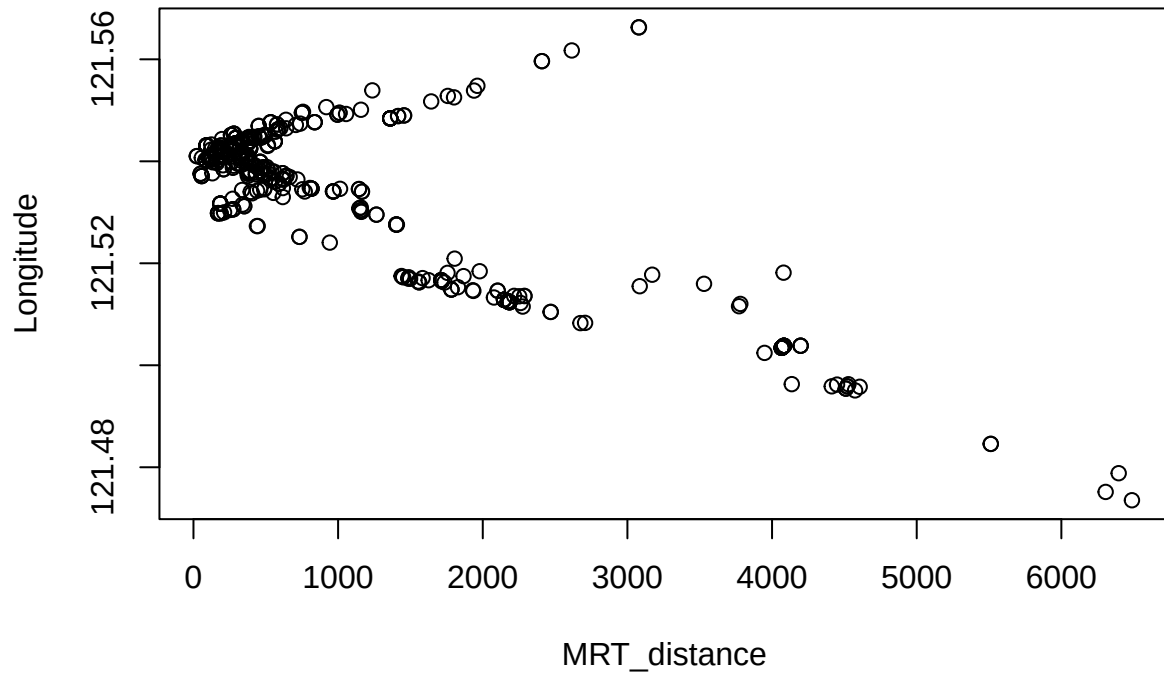




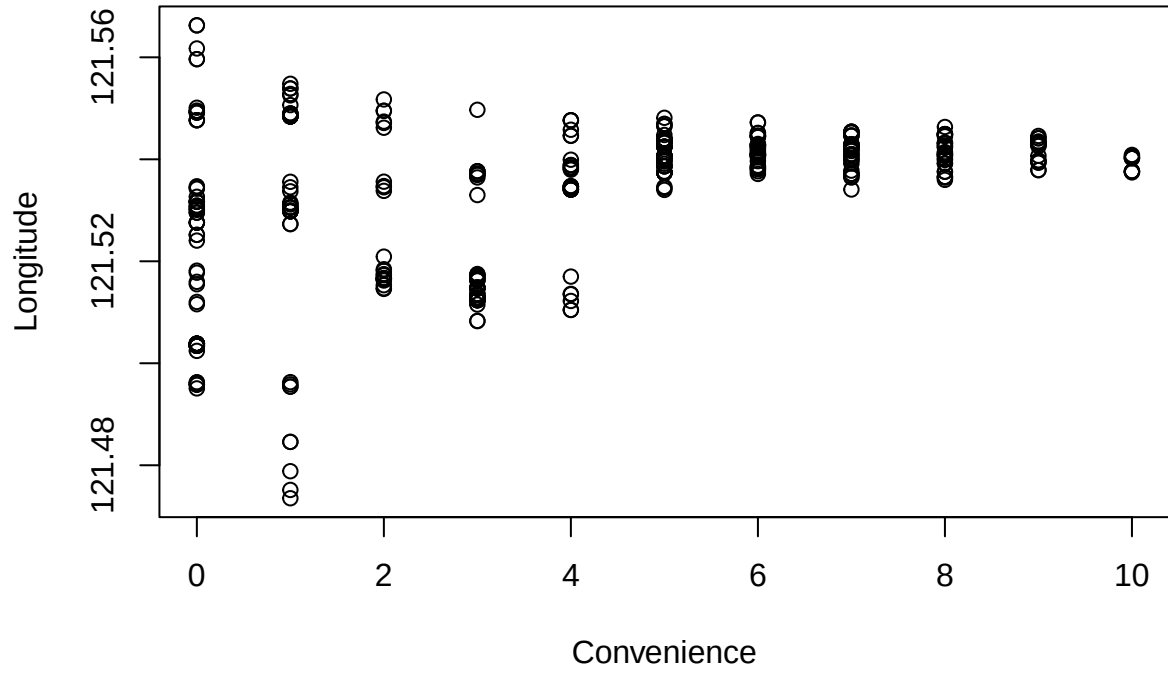
Age against Longitude



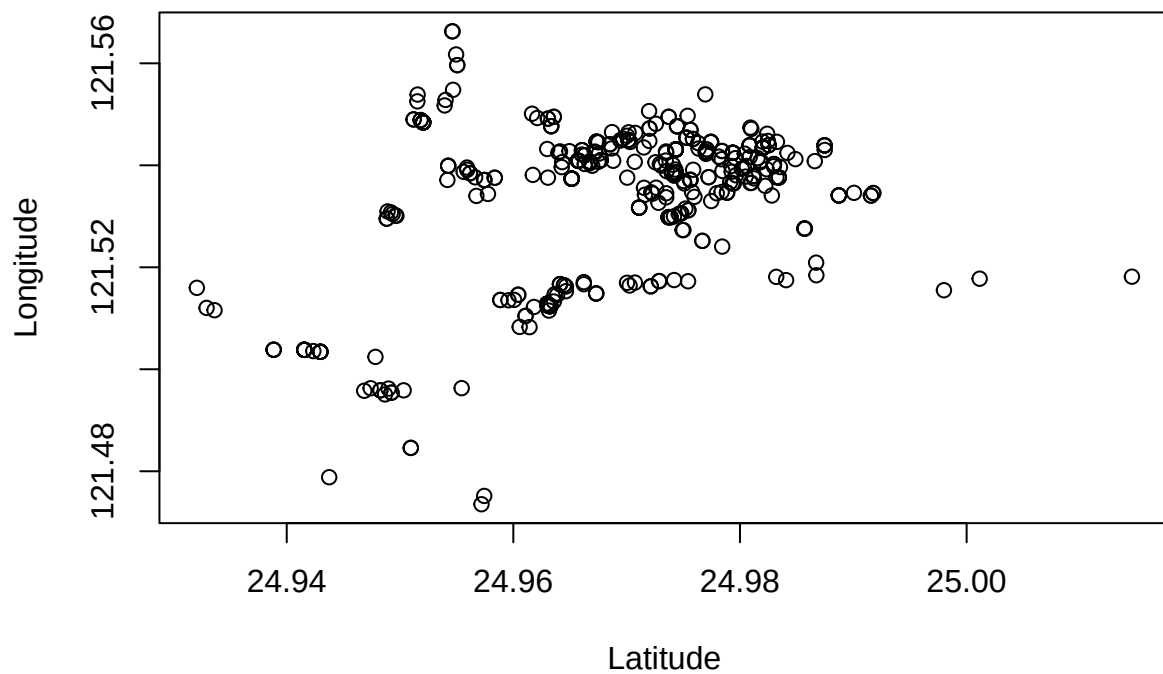
MRT_distance against Longitude



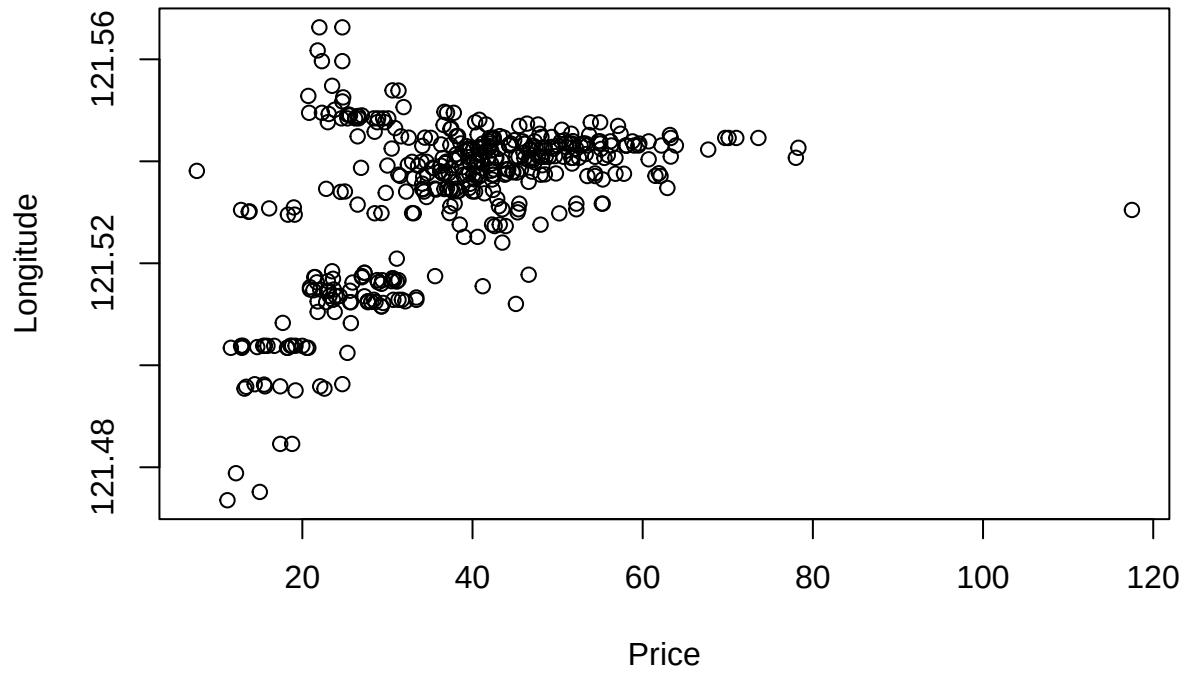
Convenience against Longitude

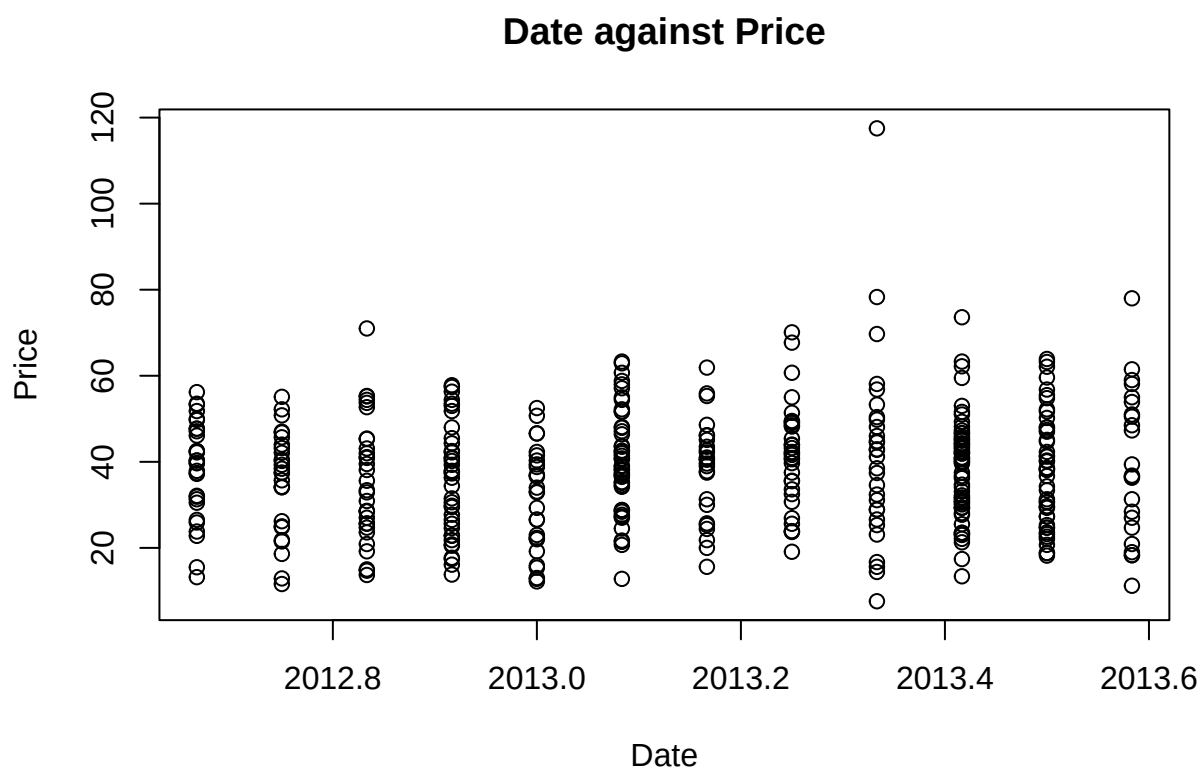


Latitude against Longitude

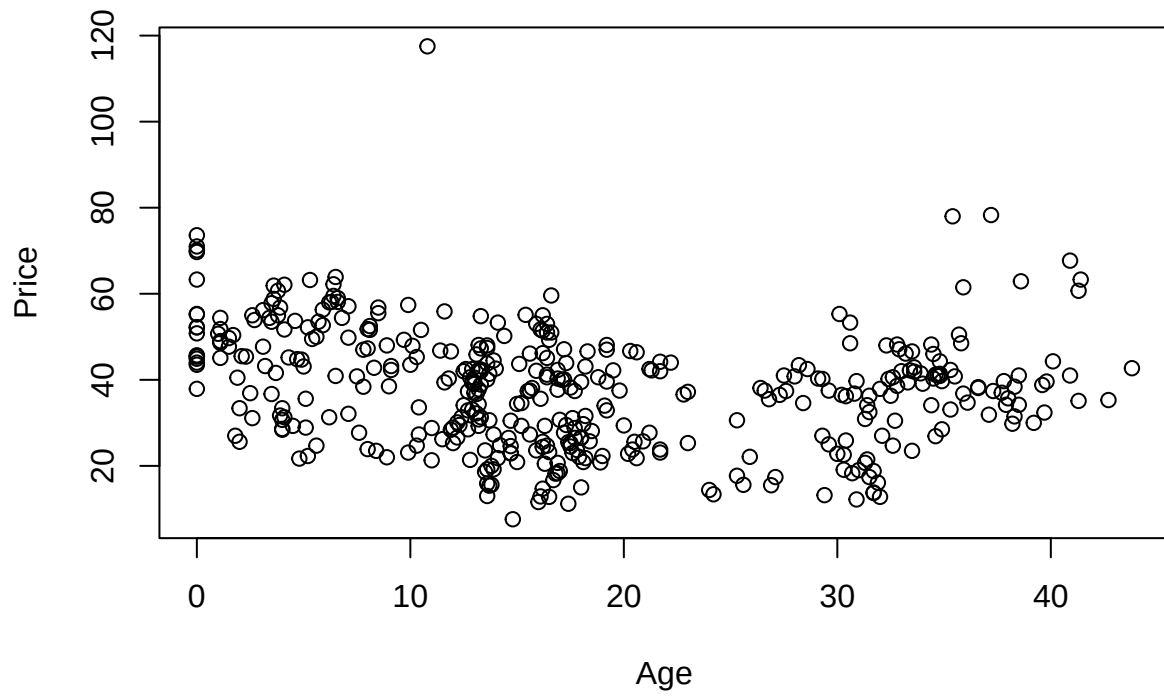


Price against Longitude

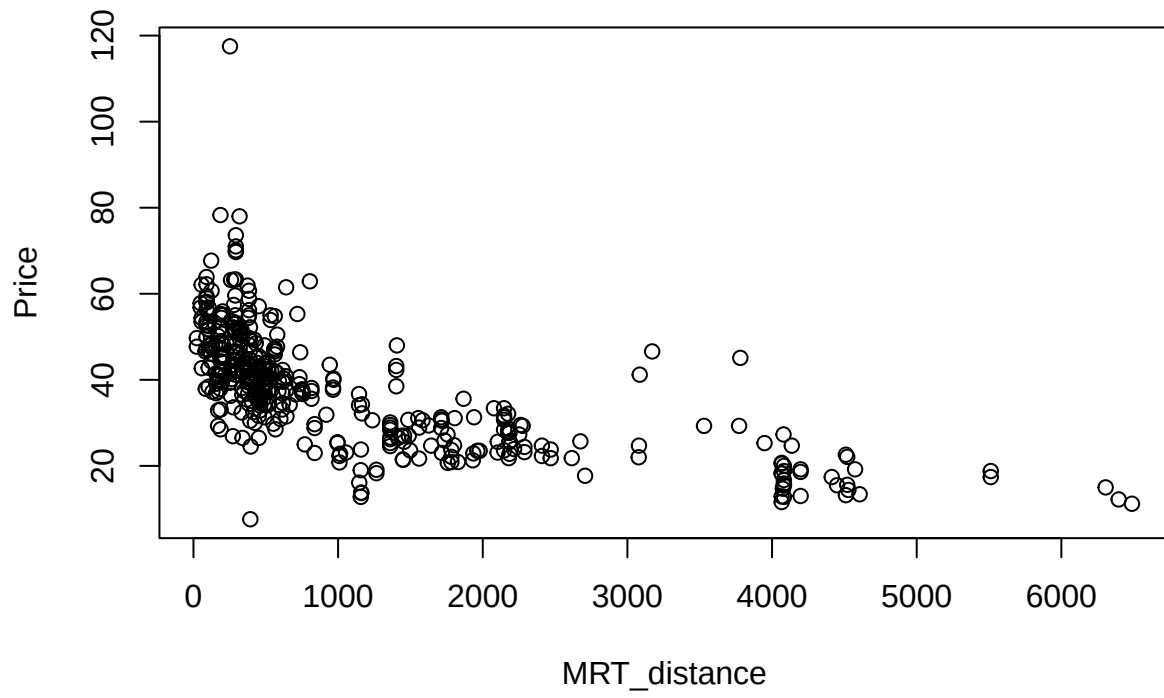




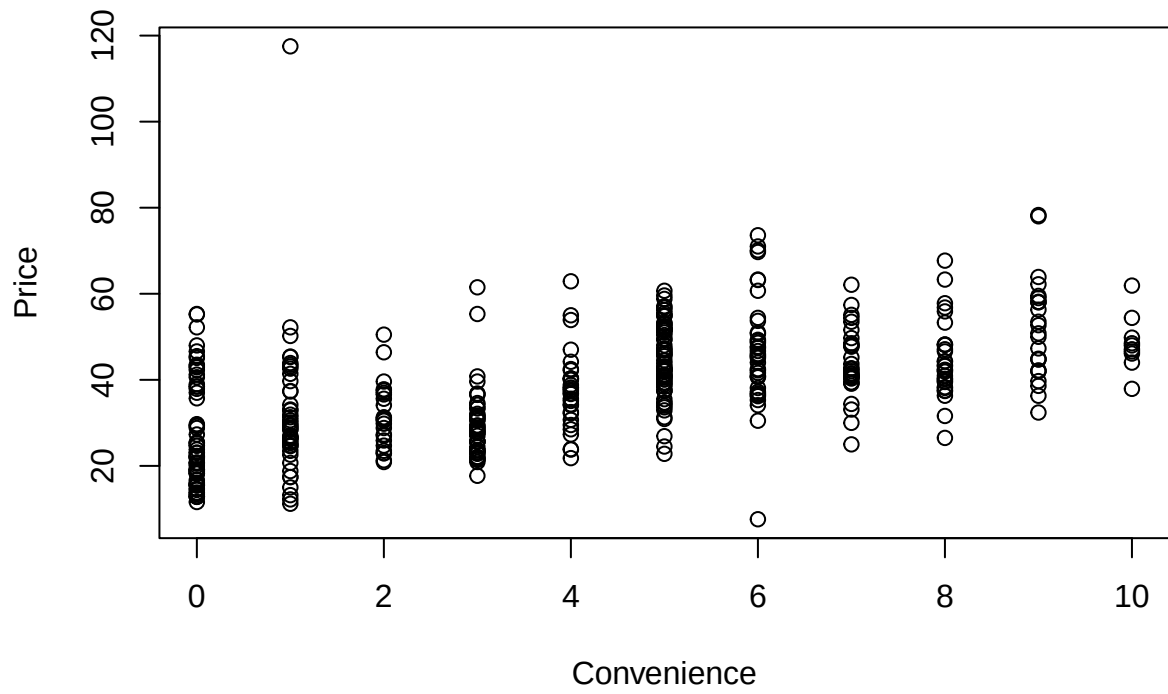
Age against Price

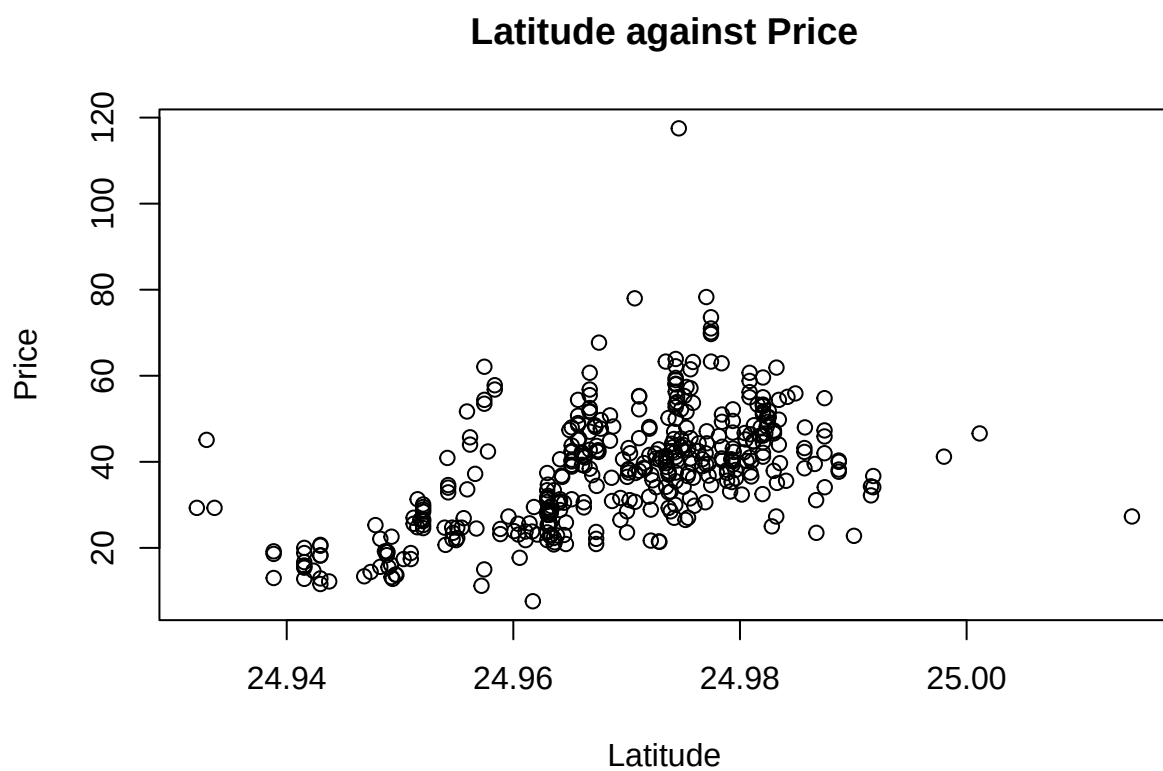


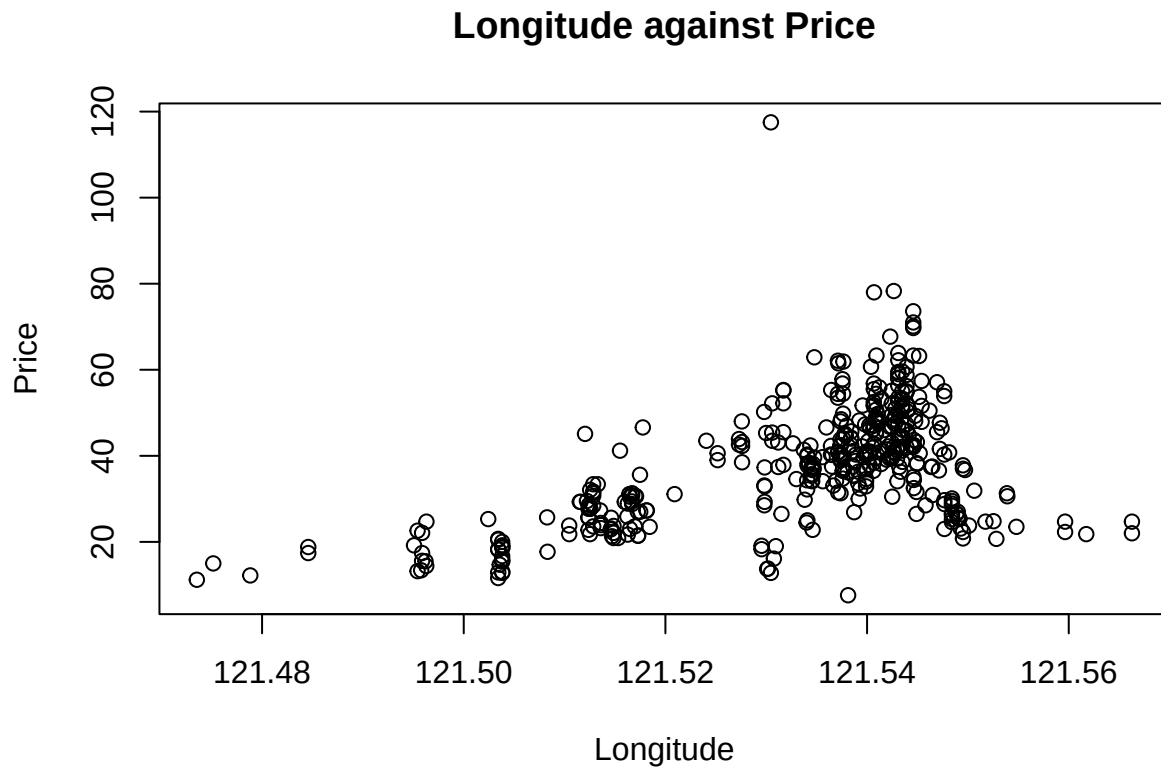
MRT_distance against Price



Convenience against Price







Question Two

```
##
## Call:
## lm(formula = Price ~ Date, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -31.159 -10.082   0.923   8.528  78.741
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -8464.260   4767.178  -1.776   0.0765 .
## Date           4.223     2.368    1.783   0.0752 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 13.57 on 412 degrees of freedom
## Multiple R-squared:  0.007661,    Adjusted R-squared:  0.005253
## F-statistic: 3.181 on 1 and 412 DF,  p-value: 0.07524
##
## Call:
## lm(formula = Price ~ Date + Age, data = data)
##
## Residuals:
```

```

##      Min      1Q  Median      3Q      Max
## -31.930 -10.951   1.643   8.248  76.956
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -8.821e+03  4.664e+03  -1.891   0.0593 .
## Date         4.403e+00  2.317e+00   1.900   0.0581 .
## Age          -2.534e-01  5.735e-02  -4.419  1.27e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 13.28 on 411 degrees of freedom
## Multiple R-squared:  0.05266,    Adjusted R-squared:  0.04805
## F-statistic: 11.42 on 2 and 411 DF,  p-value: 1.485e-05
##
## Call:
## lm(formula = Price ~ Date + Age + MRT_distance, data = data)
##
## Residuals:
##      Min      1Q  Median      3Q      Max
## -37.274  -5.393  -0.573   4.640  70.665
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.279e+04  3.370e+03  -3.795 0.000170 ***
## Date         6.377e+00  1.674e+00   3.809 0.000161 ***
## Age          -2.336e-01  4.137e-02  -5.646 3.08e-08 ***
## MRT_distance -7.295e-03  3.741e-04 -19.502 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 9.574 on 410 degrees of freedom
## Multiple R-squared:  0.5085, Adjusted R-squared:  0.5049
## F-statistic: 141.4 on 3 and 410 DF,  p-value: < 2.2e-16
##
## Call:
## lm(formula = Price ~ Date + Age + MRT_distance + Convenience,
##     data = data)
##
## Residuals:
##      Min      1Q  Median      3Q      Max
## -38.392  -5.630  -0.985   4.304  76.004
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.159e+04  3.214e+03  -3.607 0.000348 ***
## Date         5.780e+00  1.597e+00   3.620 0.000331 ***
## Age          -2.545e-01  3.953e-02  -6.438 3.40e-10 ***
## MRT_distance -5.513e-03  4.480e-04 -12.305 < 2e-16 ***
## Convenience  1.258e+00  1.918e-01   6.559 1.64e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##

```

```

## Residual standard error: 9.118 on 409 degrees of freedom
## Multiple R-squared:  0.5553, Adjusted R-squared:  0.551
## F-statistic: 127.7 on 4 and 409 DF,  p-value: < 2.2e-16

##
## Call:
## lm(formula = Price ~ Date + Age + MRT_distance + Convenience +
##     Latitude, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -35.625  -5.373  -1.020   4.243   75.343
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.596e+04  3.233e+03  -4.938 1.15e-06 ***
## Date         5.138e+00  1.554e+00   3.305 0.00103 **
## Age        -2.694e-01  3.847e-02  -7.003 1.04e-11 ***
## MRT_distance -4.353e-03  4.899e-04  -8.887 < 2e-16 ***
## Convenience  1.136e+00  1.876e-01   6.056 3.17e-09 ***
## Latitude     2.269e+02  4.417e+01   5.136 4.35e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8.847 on 408 degrees of freedom
## Multiple R-squared:  0.5823, Adjusted R-squared:  0.5772
## F-statistic: 113.8 on 5 and 408 DF,  p-value: < 2.2e-16

##
## Call:
## lm(formula = Price ~ Date + Age + MRT_distance + Convenience +
##     Latitude + Longitude, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -35.667  -5.412  -0.967   4.217   75.190
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.444e+04  6.775e+03  -2.132 0.03364 *
## Date         5.149e+00  1.557e+00   3.307 0.00103 **
## Age        -2.697e-01  3.853e-02  -7.000 1.06e-11 ***
## MRT_distance -4.488e-03  7.180e-04  -6.250 1.04e-09 ***
## Convenience  1.133e+00  1.882e-01   6.023 3.83e-09 ***
## Latitude     2.255e+02  4.457e+01   5.059 6.38e-07 ***
## Longitude    -1.243e+01  4.858e+01  -0.256 0.79820
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8.858 on 407 degrees of freedom
## Multiple R-squared:  0.5824, Adjusted R-squared:  0.5762
## F-statistic:  94.6 on 6 and 407 DF,  p-value: < 2.2e-16

```

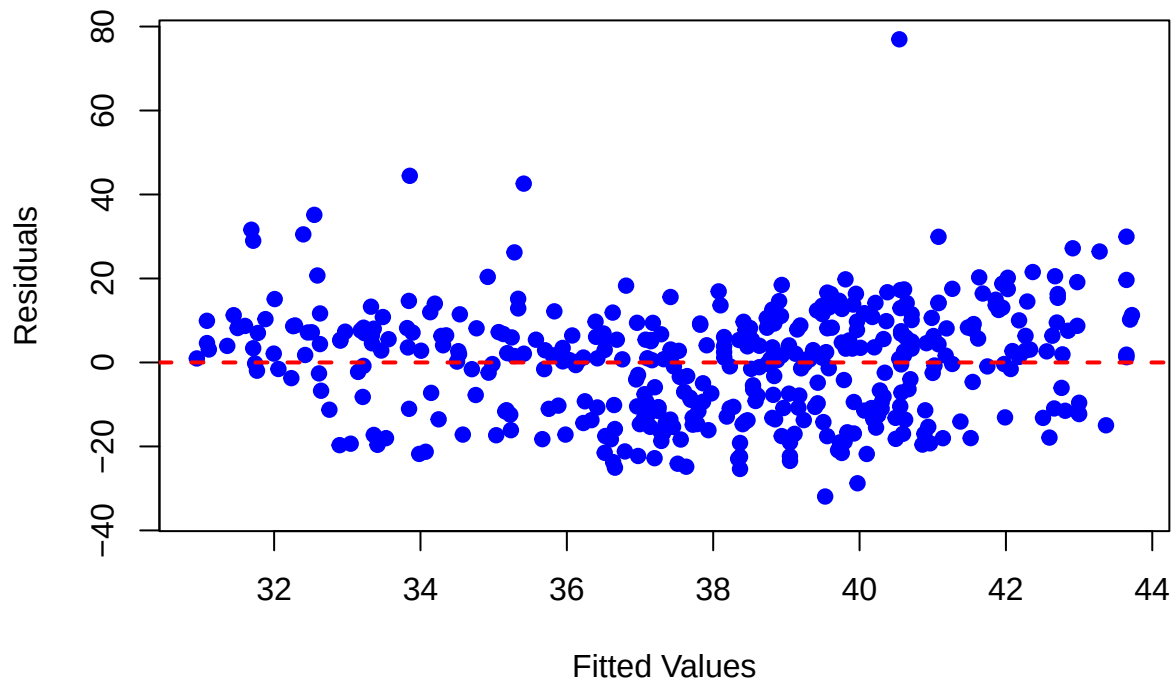
Question Three

```
## Start:  AIC=2162.53
## Price ~ 1
##
##           Df Sum of Sq  RSS    AIC
## + MRT_distance  1      34695 41767 1914.2
## + Convenience  1      24930 51531 2001.2
## + Latitude     1      22820 53641 2017.8
## + Longitude    1      20937 55524 2032.1
## + Age          1       3390 73071 2145.8
## + Date         1        586 75876 2161.3
## <none>                76461 2162.5
##
## Step:  AIC=1914.19
## Price ~ MRT_distance
##
##           Df Sum of Sq  RSS    AIC
## + Convenience  1      3273.6 38493 1882.4
## + Age          1      2859.1 38908 1886.8
## + Latitude     1      2579.5 39187 1889.8
## + Date         1      1268.0 40499 1903.4
## <none>                41767 1914.2
## + Longitude    1        86.2 41681 1915.3
##
## Step:  AIC=1882.4
## Price ~ MRT_distance + Convenience
##
##           Df Sum of Sq  RSS    AIC
## + Age          1      3402.1 35091 1846.1
## + Latitude     1      1881.3 36612 1863.7
## + Date         1      1046.1 37447 1873.0
## <none>                38493 1882.4
## + Longitude    1        23.5 38470 1884.2
##
## Step:  AIC=1846.09
## Price ~ MRT_distance + Convenience + Age
##
##           Df Sum of Sq  RSS    AIC
## + Latitude     1      2299.34 32792 1820.0
## + Date         1      1089.49 34001 1835.0
## <none>                35091 1846.1
## + Longitude    1        52.53 35038 1847.5
##
## Step:  AIC=1820.03
## Price ~ MRT_distance + Convenience + Age + Latitude
##
##           Df Sum of Sq  RSS    AIC
## + Date         1      855.04 31937 1811.1
## <none>                32792 1820.0
## + Longitude    1         2.03 32790 1822.0
##
## Step:  AIC=1811.1
## Price ~ MRT_distance + Convenience + Age + Latitude + Date
```

```
##
##           Df Sum of Sq  RSS   AIC
## <none>                31937 1811.1
## + Longitude  1      5.1353 31931 1813.0
```

Question Four

Residual Plot: Price ~ Date + Age



Residual Plot: Full Model (All Variables)

